Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares

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Abstract

Background: N-of-1 trials offer a promising approach for precision medicine by evaluating treatment effects within individual patients through repeated crossover designs. Their statistical properties relative to traditional randomized controlled trials remain incompletely characterized across clinically realistic effect-size and carryover regimes. **Methods:** We conducted an ADEMP-compliant Monte Carlo simulation study comparing the statistical power of aggregated N-of-1 hybrid trials versus parallel-group RCTs for detecting treatment effects of prazosin in reducing PTSD nightmares. The N-of-1 design enrolled 35 individuals each completing 8-period crossover trials analyzed by a linear mixed-effects model with continuous-time AR(1) residuals; the RCT arm used ANCOVA at matched sample size $N = 35$. Carryover-contaminated placebo periods were excluded from the N-of-1 analysis before fitting. We ran $n_{\text{sim}} = 2{,}000$ replicates per cell, uniform across all primary and sensitivity analyses. **Results:** The aggregated N-of-1 hybrid dominated the parallel-group RCT across the full effect-size range. At a true effect of -2.0 nightmares per week, N-of-1 power was 1.000 versus 0.675 (Monte Carlo standard error, MCSE, 0.010) for the RCT; at -1.0, N-of-1 power was 0.830 (MCSE 0.008) versus 0.230 (MCSE 0.009). Bias was negligible in both designs. Coverage of nominal-95% CIs was 0.940–0.949 for the N-of-1 arm and 0.939 for the RCT arm across all effect sizes. **Conclusions:** Aggregated N-of-1 hybrid trials achieve substantially higher statistical power than parallel-group RCTs at matched sample size when within-subject carryover is managed by period exclusion. The efficiency advantage is largest in the small-to-moderate effect-size regime where parallel-group RCTs are essentially underpowered. **Keywords:** N-of-1 trials; randomized controlled trials; statistical power; crossover design; carryover effects; PTSD; prazosin; precision medicine.

Contents

1 Abstract

Background. Aggregated N-of-1 trial designs and parallel-group randomized controlled trials are the two principal methodological alternatives for evaluating treatment effects under substantial individual heterogeneity. The published literature characterizes the variance-component foundations of each design separately, but has not, to the best of our knowledge, delivered a comprehensive sample-size and power comparison across clinically realistic effect-size and carryover regimes that admits Monte Carlo standard-error reporting within the ADEMP framework of Morris, White, and Crowther [-@morrisUsingSimulationStudies2019a].

Methods. We conduct an ADEMP-compliant Monte Carlo simulation study comparing the aggregated N-of-1 hybrid design to a parallel-group RCT at matched sample size $N = 35$, across a grid of true-effect sizes $\{0, -0.5, -1.0, -2.0\}$ nightmares per week, calibrated to the prazosin/PTSD reference application. The hybrid N-of-1 arm fits an ‘nlme::lme‘ model with continuous-time AR(1) residual correlation; the parallel-group RCT arm fits a baseline-adjusted linear model (ANCOVA). Carryover-contaminated placebo periods are excluded before fitting the N-of-1 model. Each cell is run at $n_{\text{sim}} = 2{,}000$ replicates; null cells (true effect zero) provide explicit Type I error checks. Estimands include power, empirical effect-size bias, empirical SE, model SE, and nominal-95are reported with Monte Carlo standard errors. Sensitivity sweep S1 examines carryover half-lives in $\{0.5, 1, 3, 5, 7\}$ days.

Results. The aggregated N-of-1 hybrid dominated the parallel-group RCT across the entire effect-size range. At true effect -2.0, N-of-1 power was 1.000 versus parallel-group 0.675 ± 0.010; at -1.0, 0.830 ± 0.008 versus 0.230 ± 0.009; at -0.5, 0.311 ± 0.010 versus 0.104 ± 0.007. The two power curves did not cross within the simulated range; gaps exceeded five combined MCSEs at every non-null effect size. The relative N-of-1-to-RCT power ratio was approximately $3.0\times$ at the moderate-effect -0.5 cell. Bias was negligible across all

cells for both designs. Coverage of nominal-95the RCT arm, each within two MCSEs of the nominal level. The N-of-1 analysis was insensitive to the true carryover half-life, confirming robustness of the period-exclusion strategy. Type I error was 0.051 ± 0.005 for the N-of-1 hybrid and 0.061 ± 0.005 for the RCT; the RCT rate is marginally above nominal, consistent with small-sample effects in lm at $N = 35$.

Conclusions. Aggregated N-of-1 hybrid trials achieve substantially higher statistical power than parallel-group RCTs at matched sample size when individual variability is substantial and within-subject carryover is manageable. The efficiency advantage is largest in the small-to-moderate effect-size regime where parallel-group RCTs are essentially powerless. Trial designers in precision-medicine settings with high between-patient variability should consider the aggregated N-of-1 design as the preferred default.

2 Introduction

[1]

N-of-1 trials have emerged as a methodological complement to RCTs for settings where between-patient variability is clinically central.

- Traditional RCTs establish population-level effects but do not capture between-patient variability in treatment response.
- Precision medicine approaches are motivated by substantial between-patient variance in treatment response.
- The statistical properties of N-of-1 designs relative to RCTs across realistic effect-size and carryover regimes remain incompletely characterized.

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Clinical research has moved steadily toward precision-medicine approaches that recognize substantial between-patient variance in treatment response[1,2]. Traditional randomized controlled trials (RCTs), while remaining the standard for establishing population-level treatment effects, may not capture the between-patient variability that is central to precision-medicine therapeutic decision-making[3,4].

[2]

Aggregated N-of-1 trials reduce between-patient confounding while supporting both individual and population-level treatment-effect inference.

- Each patient serves as their own control in repeated crossover periods, eliminating between-patient variability from the estimator.

- Aggregation across patients enables population-level inference while simultaneously informing individual treatment decisions.
- The design provides individual-level treatment-effect estimates unavailable from parallel-group comparisons.

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N-of-1 trials, also known as single-patient crossover trials, evaluate treatment efficacy within individual patients through repeated crossover periods[5,6]. In these designs each patient serves as their own control, which reduces the confounding effect of between-patient variability while providing individual-level treatment-effect estimates[7,8]. When appropriately aggregated across patients, N-of-1 trials can support population-level inference while simultaneously informing individual treatment decisions[9,10].

2.1 Methodological Considerations in N-of-1 Trial Design

[3]

The statistical analysis of N-of-1 trials requires explicit attention to carryover, period effects, and serial correlation within patients.

- Carryover effects from prior treatment periods are particularly critical in crossover designs and can affect validity and power.
- Temporal trends and within-patient autocorrelation present analytical challenges absent from parallel-group designs.
- Appropriate handling of these nuisance features is a prerequisite for unbiased estimation.

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The statistical analysis of N-of-1 trials presents methodological challenges that differ substantially from those of traditional parallel-group designs[11,12]. Among the key considerations are the appropriate handling of carryover effects, period effects, and temporal trends within individual patients[13,14]. Carryover effects, representing the residual influence of treatment from previous periods, are particularly critical in crossover designs and can substantially affect both the validity and the power of statistical analyses[15,16].

[4]

Population-level inference from N-of-1 series requires meta-analytic methods that account for both within- and between-patient variability.

- Random-effects meta-analysis methods such as DerSimonian-Laird appropriately combine individual treatment-effect estimates.
- Both within-patient and between-patient variance components must be incorporated to avoid bias.
- The aggregation strategy chosen influences the efficiency and interpretability of the combined estimate.

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The aggregation of individual N-of-1 trials to yield population-level estimates requires meta-analytic approaches that account for both within-patient and between-patient variability[7]. Random-effects meta-analysis methods, such as the DerSimonian-Laird approach, can appropriately combine individual treatment-effect estimates while incorporating heterogeneity between patients[17].

2.2 Sample Size and Power Considerations

[5]

The statistical efficiency of N-of-1 trials relative to RCTs is context-dependent and has not been comprehensively evaluated across effect sizes and carryover scenarios.

- Sample size in N-of-1 designs involves the joint optimization of patient count, crossover periods, and within-to-between variability ratio.
- Prior simulation studies suggest N-of-1 designs achieve adequate power at smaller total sample sizes, particularly for moderate effects with substantial individual variability.
- The relative efficiency advantage and its dependence on carryover regime remain open empirical questions.

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Sample size determination for N-of-1 trials involves the number of patients, the number of crossover periods per patient, and the expected magnitude of within-patient relative to between-patient variability[18]. Simulation studies have suggested that N-of-1 designs may achieve adequate statistical power with smaller total sample sizes than parallel-group RCTs, particularly when treatment effects are moderately large and individual variability is substantial[19]. But how large is the advantage, and how does it depend on the carryover regime? The relative statistical efficiency of N-of-1 trials compared to RCTs remains context-dependent and has not been comprehensively evaluated across varying effect sizes and carryover scenarios[9], and understanding these relationships is important for informed study design selection.

2.2.1 Analytic Frameworks for Power in N-of-1 Designs

[6]

N-of-1 statistical power depends on a fundamentally different variance decomposition than parallel-group RCTs, as between-patient variance is eliminated from the treatment-effect estimator.

- The relevant uncertainty sources are the within-patient between-period variance and residual measurement error.
- This variance reduction relative to RCTs underpins the N-of-1 efficiency advantage.
- The decomposition implies that power depends on within-patient noise and the number of crossover cycles, not between-patient heterogeneity.

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The statistical power of an N-of-1 trial depends on a fundamentally different set of variance components than a parallel-group RCT. Because each patient serves as their own control, the between-patient variance $\sigma^2_{\text{between}}$ is eliminated from the treatment-effect estimator, and the relevant sources of uncertainty become the within-patient, between-period variance σ^2 and the residual measurement error[11,20].

2.2.1.1 Variance decomposition and the paired-cycles model

[7]

Senn formalized sample-size planning for N-of-1 series under a Normal-Normal mixture model where cycle-level differences follow a random-effects structure.

- The model decomposes within-cycle treatment differences into a patient-level random effect and within-patient residual noise.
- τ^2 captures between-patient variance in individual treatment effects and $2\sigma^2$ the within-patient within-cycle variance.
- The paired-cycles structure implies that power depends on the ratio of within-patient noise to patient heterogeneity.

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Senn[20] formalized the sample size problem for sets of N-of-1 trials under a Normal-Normal mixture model. Suppose n patients are each randomized in k cycles of paired treatment periods, each cycle containing one period of treatment A and one of treatment B in random order, and let d_{ij} denote the within-cycle treatment difference for patient i in cycle j. The model assumes

$$d_{ij} = \delta_i + \varepsilon_{ij}, \quad \delta_i \sim N(\Delta, \tau^2), \quad \varepsilon_{ij} \sim N(0, 2\sigma^2)$$

[8]

The factor of 2 in the variance arises because each within-cycle difference is computed from two independent within-period observations.

- Δ is the population mean treatment effect and τ^2 is the between-patient variance of individual treatment effects.
- $2\sigma^2$ is the within-patient within-cycle variance of paired differences.
- This parameterization identifies the two variance components that determine sample size requirements.

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where Δ is the population mean treatment effect, τ^2 is the between-patient variance of individual treatment effects (the treatment-by-patient interaction), and $2\sigma^2$ is the within-patient, within-cycle variance of the paired differences[20]. The factor of 2 arises because d_{ij} is the difference of two independent within-period observations, each with variance σ^2.

Under this model the patient-level mean difference $\bar{d}_{i\cdot} = k^{-1} \sum_j d_{ij}$ has variance $\tau^2 + 2\sigma^2/k$, and the grand mean $\hat{\Delta} = n^{-1} \sum_i \bar{d}_{i\cdot}$ has variance

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n}$$

[9]

The grand mean estimator variance $\mathrm{Var}(\hat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$ reveals the trade-off between number of patients and cycles per patient.

- Increasing k reduces only the within-patient $2\sigma^2/k$ component, while increasing n reduces the entire expression.
- When between-patient heterogeneity τ^2 dominates, additional patients yield greater precision than additional cycles.
- This trade-off is central to sample-size planning for N-of-1 series and has no simple universal optimum.

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This expression is central to the trade-off between the number of patients n and the number of cycles per patient k: increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally[7,20].

2.2.1.2 Closed-form power for fixed- and random-effects analyses

[10]

Senn distinguished random-effects and fixed-effects inferential goals with distinct power formulas and degrees-of-freedom structures.

- The random-effects analysis targets the population treatment effect Δ and yields a one-sample t-test on the n patient-level means.
- The noncentrality parameter for the random-effects case is $\lambda_{\mathrm{RE}} = \Delta/\sqrt{(\tau^2 + 2\sigma^2/k)/n}$ with $n-1$ degrees of freedom.
- The random-effects analysis estimates the patient-level variance directly from the n observed summary measures.

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Senn[20] distinguished two inferential goals with distinct power formulas. For a *random-effects* analysis targeting the population treatment effect Δ, the patient-level summary-measures approach yields a one-sample t-test on the n patient means $\bar{d}_{i\cdot}$ with $n-1$ degrees of freedom; the variance at the patient level is $\tau^2 + 2\sigma^2/k$, estimated directly from the n observed summary measures. Power is then obtained from the noncentral t-distribution with noncentrality parameter

$$\lambda_{\mathrm{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}$$

and $n-1$ degrees of freedom.

[11]

The fixed-effects analysis excludes the treatment-by-patient interaction from the variance estimator and uses $n(k-1)$ degrees of freedom from pooled within-patient sums of squares.

- The relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$, estimated with $n(k-1)$ degrees of freedom.
- The noncentrality parameter is $\lambda_{\mathrm{FE}} = \Delta/\sqrt{2\sigma^2/(nk)}$.
- Standard sample size software assuming $n-1$ degrees of freedom slightly underestimates power in the fixed-effects case when $k > 2$.

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For a *fixed-effects* analysis, appropriate when the inferential goal is to test whether the treatments differ for the patients actually studied, the treatment-by-patient interaction τ^2 is excluded from the variance estimator. That is, the relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$, estimated with $n(k-1)$ degrees of freedom obtained by pooling within-patient sums of squares. Power follows from the noncentral t-distribution with

$$\lambda_{\mathrm{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}$$

and $n(k-1)$ degrees of freedom[20]. Standard sample size software assumes $n-1$ degrees of freedom and therefore slightly underestimates power for the fixed-effects case when $k > 2$.

[12]

Senn and Julious provided a complementary tutorial on computing pooled within-patient standard deviations and constructing shrinkage estimators for individual patient effects.

- The pooled within-patient SD is estimated as $\hat{\sigma}_w = \sqrt{\sum_i \mathrm{SS}_i / \sum_i (m_i - 1)}$.
- Shrinkage estimators, that is best linear unbiased predictors (BLUPs), for individual treatment effects can be derived from the random-effects model.
- These practical tools bridge the theoretical variance decomposition and applied N-of-1 analysis.

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Senn and Julious[12] provided a complementary tutorial demonstrating how these paired-cycle differences are computed and analyzed in practice, including the estimation of the pooled within-patient standard deviation $\hat{\sigma}_w = \sqrt{\sum_i \mathrm{SS}_i / \sum_i (m_i - 1)}$ and the construction of shrinkage (BLUP) estimators for individual patient effects.

2.2.1.3 The n-versus-k trade-off in series of N-of-1 trials

[13]

Zucker et al. showed that the marginal benefit of additional crossover cycles diminishes rapidly once within-patient noise falls below between-patient heterogeneity.

- When $\tau^2 \gg \sigma_w^2$, additional patients yield greater precision than additional cycles.

- Conversely, when within-patient noise is large relative to heterogeneity, repeated cycles on the same patient remain valuable.
- This result motivates joint optimization of n and k in N-of-1 series planning.

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Zucker et al.[7] studied the trade-off between the number of patients n and the number of paired treatment periods k using a random-effects meta-analysis framework. Their precision formula,

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}$$

where σ_w^2 denotes the within-patient variance of paired differences, shows that the marginal benefit of additional cycles diminishes rapidly once $\sigma_w^2/k \ll \tau^2$. When between-patient heterogeneity dominates (that is, $\tau^2 \gg \sigma_w^2$), additional patients yield greater precision gains than additional cycles; conversely, when within-patient noise is large relative to heterogeneity, repeated cycles on the same patient remain valuable.

[14]

Arends et al. extended Zucker et al.'s results to derive explicit sample-size formulas for N-of-1 series analyzed via random-effects meta-analysis.

- Explicit tables jointly determine n and k as a function of true effect Δ, within-patient variance σ_w^2, and between-patient heterogeneity τ^2.
- These formulas provide a closed-form foundation for N-of-1 trial planning in the absence of carryover or serial correlation.
- The present simulation study tests the closed-form predictions under realistic violations of the simplifying assumptions.

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Arends et al.[18] extended these results to derive explicit sample-size formulas for series of N-of-1 trials analyzed via random-effects meta-analysis, providing tables for jointly determining n and k as a function of Δ, σ_w^2, and τ^2.

2.2.1.4 Complications motivating simulation

[15]

Closed-form power solutions assume negligible carryover, an assumption that may not hold when pharmacokinetic

washout between periods is incomplete.

- Partial carryover attenuates the effective treatment contrast and shifts variance components in ways not captured by balanced-design formulas.
- Prazosin's neuroadaptive effects on sleep architecture may produce precisely such partial washout.
- These violations motivate Monte Carlo simulation as a complement to closed-form sample-size planning.

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The closed-form solutions above rest on several simplifying assumptions that may not hold in practice. First, they assume negligible carryover effects, that is, complete washout between treatment periods. When carryover is partial, as may occur with prazosin's neuroadaptive effects on sleep architecture, the effective treatment contrast is attenuated and variance components shift in ways not captured by balanced-design formulas[14].

[16]

Ignoring within-patient serial correlation can inflate Type I error or distort power estimates in N-of-1 trials.

- Wang and Schork demonstrated via AR(1) likelihood-ratio methods that serial correlation can substantially reduce power.
- More frequent period alternation partially mitigates this power loss at the cost of practical feasibility.
- Designs with few, long treatment periods are most vulnerable to power loss from unmodeled serial correlation.

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Second, within-patient observations are typically serially correlated, and ignoring this correlation can inflate Type I error or distort power estimates[13]. Wang and Schork[21] demonstrated via likelihood-ratio methods under an AR(1) correlation structure that serial correlation can substantially reduce power, particularly in designs with few, long treatment periods, and they showed that more frequent period alternation mitigates this power loss at the cost of practical feasibility.

[17]

Several additional violations of the balanced paired-cycles assumptions further motivate Monte Carlo simulation for realistic N-of-1 power evaluation.

- Practical N-of-1 trials may feature unbalanced designs, missing periods, or adaptive stopping rules.

- Count data outcomes violate the normality assumption underlying closed-form formulas.
- Joint optimization of n and k for designs targeting both individual and population-level inference has no simple closed form.

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Third, practical N-of-1 trials may feature unbalanced designs with unequal period lengths, missing periods, or adaptive stopping rules that violate the balanced paired-cycles assumption. Fourth, the outcome of interest (e.g., nightmare frequency) may be better modeled as count data, whereas the closed-form formulas assume normality. Fifth, when the inferential goal encompasses both individual-level treatment decisions and population-level inference, the power trade-off between n and k has no simple closed-form solution because both the shrinkage estimator and the population estimate must be jointly optimized[20].

[18]

These accumulated complications collectively motivate the Monte Carlo simulation approach, which permits power evaluation under realistic combinations of carryover, variance structure, and design imbalance.

- Simulation accommodates carryover, serial correlation, and design imbalance simultaneously.
- ADEMP-compliant simulation reporting with Monte Carlo standard errors provides quantified uncertainty for all performance measures.
- The present study fills the gap left by closed-form analyses that assume idealized conditions.

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These complications collectively motivate the Monte Carlo simulation approach adopted in the present study, which permits evaluation of power under realistic combinations of carryover magnitude, variance structure, and design imbalance.

2.3 Single-Case Experimental Design Framework

[19]

N-of-1 trials represent a specific application of single-case experimental designs whose methodological rigor has evolved to support valid causal inferences from individual-level data.

- SCED methodology has been extensively developed in behavioral and educational research contexts.
- Analytical approaches for SCEDs now include tools for valid causal inference from individual trajectories.
- Methodological standardization from the SCED literature provides a foundation for rigorous N-of-1 trial analysis.

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N-of-1 trials represent a specific application of single-case experimental designs (SCEDs) that have been extensively developed in behavioral and educational research[22,23]. The methodological rigor of SCEDs has evolved to include analytical approaches that can provide valid causal inferences from individual-level data[24,25].

[20]

Recent SCED advances emphasizing proper randomization, adequate baseline measurement, and appropriate statistical analysis have strengthened the evidence base for N-of-1 trials as a complement to group designs.

- Randomization, adequate baseline, and appropriate analysis are now recognized as essential SCED quality criteria.
- These advances position N-of-1 trials as methodologically credible complements to parallel-group designs.
- Standardised SCED reporting and analysis guidelines reduce implementation barriers.

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Recent developments in SCED methodology have emphasized the importance of proper randomization, adequate baseline measurement, and appropriate statistical analysis[26]. These advances have strengthened the evidence base for using N-of-1 trials as a complement to traditional group-based designs[27].

2.4 Clinical Application: Prazosin for PTSD

[21]

Prazosin for PTSD nightmares provides an instructive clinical context for N-of-1 methodology given substantial individual response variation and the chronic fluctuating nature of PTSD symptoms.

- Prazosin has demonstrated efficacy in multiple RCTs for reducing trauma-related nightmares via α_1-adrenergic blockade.

- Individual response patterns vary considerably, making individual-level inference an important clinical goal.
- The PTSD/nightmare context provides empirically grounded simulation parameter calibration.

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Post-traumatic stress disorder (PTSD) represents a particularly instructive clinical context for evaluating N-of-1 trial methodology, given the substantial individual variation in treatment response and the chronic, fluctuating nature of symptoms[28]. Prazosin, an α_1-adrenergic receptor antagonist, has demonstrated efficacy for reducing trauma-related nightmares in multiple RCTs[29,30], though individual response patterns vary considerably.

[22]

Prazosin's pharmacokinetics, including its relatively short half-life, make it suitable for crossover designs while necessitating careful consideration of carryover effects.

- The short half-life facilitates crossover trial design by limiting washout duration.
- Previous studies document both efficacy and substantial individual variability of prazosin response.
- This empirical foundation justifies the simulation parameters employed in the present study.

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The pharmacokinetic properties of prazosin, including its relatively short half-life, make it particularly suitable for crossover designs while also necessitating careful consideration of carryover effects[31,32]. Previous studies have documented both the efficacy and the individual variability of prazosin response[33,34], providing an empirical foundation for the simulation parameters we employ here.

2.5 Regulatory and Implementation Considerations

[23]

Increasing regulatory recognition of N-of-1 trials has highlighted the need for robust methodological guidelines and statistical frameworks.

- Regulatory agencies and research institutions have acknowledged N-of-1 trials as a valid evidence-generation approach.
- Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation.

- Specialised software tools and reporting standards have emerged to support N-of-1 trial analysis.

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The increasing recognition of N-of-1 trials by regulatory agencies[35] and research institutions[2] has highlighted the need for robust methodological guidelines and statistical frameworks. Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation[6,36], while specialized software tools have emerged to support N-of-1 trial analysis[25,37].

2.6 Prior Simulation Comparisons of N-of-1 and Parallel Designs

[24]

The comparative efficiency of N-of-1 and parallel-group designs has been examined in several simulation studies that form the immediate methodological precedent for the present work.

- A companion literature review provides a structured summary of the key findings.
- Direct head-to-head simulation comparisons, analytical complications, closed-form frameworks, and recent design refinements are each addressed.
- Key findings from prior studies are summarized in the following subsections.

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The comparative efficiency of N-of-1 and parallel-group designs has been examined in several simulation studies that form the immediate methodological precedent for the present work. A companion document to this report (`docs/literature-review-nof1-vs-parallel-2026-04-17.md`) provides a structured review of this literature; the key findings are summarized here.

2.6.1 Direct head-to-head simulation comparisons

[25]

Blackston et al. reported that aggregated N-of-1 designs achieve higher power than both parallel RCTs and two-period crossovers at matched sample sizes, and that unmodeled carryover inflates N-of-1 Type I error.

- Using 5,000 simulated trials with three cycles, N-of-1 designs exceeded 80% power at $n = 30$ while RCTs required $n = 150$.

- Type I error under N-of-1 inflated to 15–25% when moderate-to-strong carryover was present but not modeled.
- Patient-level random effects are recoverable in N-of-1 but not in parallel RCTs, which provide only one observation per participant.

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The most frequently cited direct comparison is Blackston and colleagues' study of aggregated N-of-1 trials versus parallel and crossover RCTs[19]. Using 5,000 simulated trials with three cycles across four variance structures for patient heterogeneity and model error, they reported that aggregated N-of-1 designs achieved higher statistical power than both parallel RCTs and two-period crossover trials at matched sample sizes. They also found that Type I error could be inflated under N-of-1 when moderate-to-strong carryover or washout was present but not modeled, and that patient-level random effects were recoverable in N-of-1 but not in the parallel RCT, because the latter provides only one observation per participant.

[26]

Carrozzo et al. found that a meta-analytic N-of-1 procedure reached 80% power with fewer participants than two-arm parallel or two-period crossover designs across most heterogeneity and carryover settings.

- The simulation varied between-subject heterogeneity and carryover magnitude for a cardiovascular digital-health intervention.
- Both a fixed-n meta-analysis and a sequential-aggregation procedure were evaluated.
- The sequential variant crossed the 80% power threshold earlier than the fixed-n procedure.

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A more recent comparison by Carrozzo and colleagues[38] examined a two-arm parallel RCT, a two-period crossover, and a meta-analysis of N-of-1 trials for a cardiovascular digital-health intervention. Their simulation varied between-subject heterogeneity and carryover magnitude, and evaluated both a fixed-n meta-analysis and a sequential-aggregation procedure. They reported that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm designs across most heterogeneity and carryover settings, and that the sequential variant crossed the 80% threshold earlier than the fixed-n procedure.

[27]

Hendrickson et al.'s simulation of aggregated N-of-1 designs for predictive-biomarker validation is the methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses.

- Four designs were compared at 1,000 replicates per scenario: open label, blinded discontinuation, traditional crossover, and a hybrid.
- The hybrid design provided power close to the traditional crossover while preserving an initial open-label titration phase.
- The hybrid showed smaller power erosion under carryover than the other designs.

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The specific methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses is Hendrickson and colleagues' simulation of aggregated N-of-1 designs for predictive-biomarker validation[39]. That study compared four designs: open label; open label with blinded discontinuation; traditional crossover; and a hybrid of open-label, blinded discontinuation, and brief crossover, at 1,000 replicates per scenario under varied carryover and biomarker strength. The hybrid design provided power close to the traditional crossover while preserving an initial open-label titration phase and showing smaller power erosion under carryover.

2.6.2 Analytical complications and misspecification

[28]

Wang and Schork extended power analysis to cases with AR(1) serial correlation, heteroscedastic responses, and washout periods, demonstrating that ignoring serial correlation substantially reduces power.

- More frequent period alternation partially mitigates serial-correlation power loss at the cost of practical feasibility.
- Washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.
- These findings motivate the `corCAR1` residual correlation structure used in the present N-of-1 analysis model.

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Wang and Schork[21] extended the analytical and simulation treatment of power to cases with AR(1) serial correlation, heteroscedastic responses, multiple evaluation periods, and washout periods. They reported that ignoring serial correlation substantially reduces power, that more frequent period alternation partially mitigates this loss at the cost of feasibility, and that washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.

[29]

Gaertner et al. simulated aggregated N-of-1 trials under directed acyclic graphs encoding carryover structures, finding that a carryover-adjusted parametric model is unbiased while other methods show bias under carryover.

- Linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted model were compared.
- All methods perform well without carryover; under carryover, only the carryover-adjusted model is unbiased.
- This work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

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Gaertner and colleagues[40] simulated aggregated N-of-1 trials under directed acyclic graphs that encode carryover and treatment-dependent covariate structures. They compared linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted parametric model, and reported that all methods perform well without carryover; under carryover, the carryover-adjusted parametric model is unbiased while the other methods show bias. Their work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

2.6.3 Closed-form and semi-analytical frameworks

[30]

Senn identified five sample-size planning criteria with closed-form solutions for N-of-1 series and characterized the n-versus-k trade-off through the variance decomposition $\mathbf{Var}(\hat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$.

- The decomposition makes explicit that increasing k reduces only the within-patient $2\sigma^2/k$ term, while increasing n reduces the full expression.
- Zucker, Ruthazer, and Schmid compared summary and individual-participant-data meta-analyses on a published N-of-1 series and characterized the variance structure determining relative efficiency.
- These closed-form results provide benchmarks against which the present simulation findings can be assessed.

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Senn formalized sample-size planning for sets of N-of-1 trials under a Normal-Normal mixture model and identified five planning criteria with corresponding closed-form solutions[20]. The decomposition $\mathrm{Var}(\widehat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$ makes explicit the trade-off between the number of patients n and the number of cycles per patient k: increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally. Zucker, Ruthazer, and Schmid[41] compared summary and individual-participant-data meta-analyses, repeated-measure models, Bayesian hierarchical models, and period or pair crossover models on a published N-of-1 series, and characterized the within- and between-patient variance structure that determines relative efficiency.

2.6.4 Recent design refinements

[31]

Recent work extends N-of-1 design analysis in three directions: sequential monitoring, per-trial monitoring with random-effects combination, and generalized pairwise comparisons.

- Jiang et al. proposed sequential monitoring rules for single-person and multi-person N-of-1 trials with sample-size calculations in an Alzheimer's disease application.
- Selukar et al. found that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with alpha-spending corrections.
- Giai et al. reported that individual-level Net Benefit estimation requires very long observation series to achieve adequate power.

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Recent work extends the comparison in three directions. Jiang and colleagues[42] proposed sequential monitoring rules for both single-person and multi-person N-of-1 trials, with sample-size calculations in an Alzheimer's disease motivating application. Selukar, Prince, and May[43] proposed sequential monitoring at the per-trial level with inverse-variance random-effects combination across trials, and reported that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with more blocks or alpha-spending corrections. Giai and colleagues[44] evaluated generalized pairwise comparison (Net Benefit) estimators in N-of-1 series, and reported that individual-level Net Benefit estimation requires very long observation series to be adequately powered.

2.6.5 Summary

[32]

The N-of-1 efficiency advantage is largest when between-patient heterogeneity is non-negligible, carryover is absent or correctly modeled, and within-patient serial correlation is accounted for.

- The advantage erodes when carryover exceeds the period length, when strong AR(1) correlation is ignored, or when the number of cycles per patient is small.
- Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs.
- Biomarker-treatment interaction analyses have qualitatively different power profiles than average-effect analyses.

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Across this literature the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible, when carryover is absent or correctly modeled, and when within-patient serial correlation is accounted for or weak. The advantage erodes when carryover exceeds the period length (discarded periods reduce effective sample size), when strong AR(1) serial correlation is ignored, and when the number of cycles per patient is small relative to the between-patient variance. Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs, and interaction-target analyses (biomarker-treatment) have qualitatively different power profiles than average-effect analyses. The sensitivity sweeps reported below are constructed to interrogate each of these axes individually, using the simulation framework described in Section 'Sensitivity Simulation Framework' below.

2.7 Objectives

[33]

The primary objective is to compare statistical power of aggregated N-of-1 hybrid trials versus parallel-group RCTs, with secondary objectives examining carryover effects, period-exclusion robustness, and conditions favoring N-of-1.

- The prazosin/PTSD application provides a clinically calibrated reference for simulation parameter specification.
- Secondary objectives include evaluating carryover half-life impact and the robustness of the period-exclusion strategy.
- The conditions under which the N-of-1 approach is preferred over parallel-group designs are explicitly characterized.

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The primary objective of this study was to compare the statistical power of aggregated N-of-1 trials versus traditional parallel-group RCTs for detecting treatment effects, using prazosin for PTSD nightmares as a clinically calibrated reference application. Secondary objectives included: (1) evaluating the impact of carryover effects with varying pharmacokinetic half-lives on statistical power; (2) examining the robustness of the period-exclusion carryover-management strategy; and (3) identifying the conditions under which the N-of-1 approach may be preferred over traditional parallel-group designs.

3 Methods

[34]

We follow the ADEMP framework of Morris, White, and Crowther, organizing the simulation study around Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.

- ADEMP provides a structured reporting standard for Monte Carlo simulation studies.
- Pre-registration of the parameter grid fixes all primary and sensitivity cells before production runs.
- Each component of ADEMP is described in turn in the following subsections.

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We follow the ADEMP framework of Morris, White, and Crowther[45]: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. In this section we shall describe each component in turn.

3.1 Estimands and pre-registration

[35]

The primary estimand is the average treatment-effect coefficient β_D from the N-of-1 linear mixed-effects model, with statistical power $\pi = \Pr(p < 0.05)$ as the primary inferential output.

- Secondary estimands include the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error.

- All performance measures are reported with binomial-proportion Monte Carlo standard errors for proportions and standard MCSE for continuous quantities.
- The full ADEMP plan is pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-p`

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The **primary estimand** is the average treatment-effect coefficient β_D from the linear mixed-effects analysis-model formula `Sx ~ Dbc + t + (1|ptID)` with `corCAR1(form = ~t|ptID)` for the aggregated N-of-1 arm, and the corresponding ANCOVA endpoint coefficient for the parallel-group comparator. The **primary inferential output** is the power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$. **Secondary estimands** are the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error of $\hat{\beta}_D$ across replicates. **Performance measures** are reported with binomial-proportion Monte Carlo standard errors $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the standard MCSE of the within-replicate mean for continuous quantities. The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`, which fixes the parameter grid for the primary comparison (M1) and the twelve sensitivity sweeps (S1-S12) before the production runs.

3.2 Simulation design and data-generating mechanism

[36]

Both designs operate on the same patient pool within each replicate, following the Morris et al. paired-comparison discipline to reduce Monte Carlo variance of power differences.

- Each replicate draws $N = 35$ patient-level baseline nightmare frequencies from a Normal distribution calibrated to yield $I^2 \approx 50\%$.
- The shared pool drives the N-of-1 and RCT arms independently within each replicate.
- The within-replicate covariance between power estimates is exploited when reporting paired power differences.

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Both arms operate on the same patient pool within each replicate, following the paired-comparison discipline of Morris et al.[45(sec5.4)]. Each replicate begins by drawing $N = 35$ patient-level baseline nightmare frequencies from Normal(μ_0, σ_τ^2) with $\mu_0 = 8.0$ nightmares per week and $\sigma_\tau = 1.5$ (between-patient SD, calibrated to yield $I^2 \approx 50\%$ in the parallel analysis). This shared pool then drives the N-of-1 arm and the RCT arm independently; the within-replicate covariance between the two power estimates is exploited when reporting paired power differences.

[37]

The N-of-1 arm simulates 8-period balanced crossover trials with continuous-time AR(1) within-patient residuals and exponential carryover decay.

- Within-patient residuals follow a zero-mean multivariate normal with $\mathrm{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$.
- Carryover from a drug period decays as $\delta_c = \beta_D \exp\{-\ln(2)L/t_{1/2}\}$, with $t_{1/2} = 3$ days at the primary cell.
- The true effect grid is $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares per week.

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For the **N-of-1 arm**, each patient completes 8 periods (4 prazosin, 4 placebo) in a balanced random sequence, with each period lasting 7 days. Within-patient residuals are drawn from a zero-mean multivariate normal distribution whose covariance follows the continuous-time AR(1) structure with SD $\sigma_w = 1.0$ and decay rate $\rho = 0.3$, giving $\mathrm{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$ where t_j indexes the period number. Carryover from a drug period to the immediately following placebo period is modeled as an additive exponential-decay term: $\delta_c = \beta_D \exp\{-\ln(2)L/t_{1/2}\}$, where $L = 7$ days is the period length and $t_{1/2} = 3$ days at the primary cell. The true effect size grid is $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares per week.

[38]

The parallel-group RCT arm allocates the same 35 patients 1:1 to treatment or placebo, with the endpoint drawn from a linear model using the shared-pool baseline.

- The endpoint model is $Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i$, using the patient's baseline draw from the shared pool.
- This structure ensures both arms are exposed to the same between-patient heterogeneity.
- The ANCOVA analysis uses the shared-pool baseline as covariate.

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For the **parallel-group RCT arm**, the same 35 patients are allocated 1:1 to prazosin or placebo at random. The endpoint is a single measurement at the end of the study period: $Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i$, where u_i is the patient's baseline draw from the shared pool, Z_i is the treatment indicator, and $\varepsilon_i \sim \text{Normal}(0, \sigma_w^2)$.

[39]

Carryover sensitivity sweep S1 varies the pharmacokinetic half-life across five values at fixed true effect $\beta_D = -2.0$.

- Half-life values $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days span a clinically plausible range.
- The sweep tests the robustness of the period-exclusion strategy to half-life misspecification.
- Fixed effect size isolates the carryover half-life as the sole varying parameter.

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Carryover sensitivity sweep S1 varies $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days at fixed $\beta_D = -2.0$.

Units. Carryover half-lives $t_{1/2}$ are quoted in **days** throughout this paper, matching prazosin's pharmacokinetic scale. Companion papers in this series that target the biomarker-treatment interaction quote $t_{1/2}$ in **weeks** on the canonical grid $\{0, 0.5, 1.0\}$; the conversion is 1 week = 7 days, so that grid corresponds to $\{0, 3.5, 7\}$ days on the present scale. Timepoints t are weekly in both conventions.

3.3 Statistical analysis models

[40]

The N-of-1 arm is analyzed with a linear mixed-effects model with patient random intercept and `corCAR1` residual correlation, tested via Satterthwaite-corrected Wald t-statistic.

- The formula `Sx ~ Dbc + t_wk` with `corCAR1(form = ~t_wk|ptID)` fits continuous-time AR(1) residual correlation.
- The Satterthwaite denominator degrees of freedom correct the small-sample anti-conservatism observed in preliminary runs at $N = 35$.
- The null hypothesis $H_0 : \beta_D = 0$ is tested at the two-sided $\alpha = 0.05$ level.

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N-of-1 arm. Each replicate is analyzed with the linear mixed-effects model

$$Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \beta_t t_{ij} + u_i + \varepsilon_{ij}$$

where Y_{ij} is the observed symptom score (code column `Sx`), $D_{bc,ij}$ is the continuous exposure-decayed drug indicator (code column `Dbc`), t_{ij} is the within-patient period index, $u_i \sim \text{Normal}(0, \sigma_u^2)$ is the patient random intercept, and ε_i has `corCAR1` covariance. In R: `nlme::lme(Sx ~ Dbc + t_wk, random = ~1|ptID, correlation = corCAR1(form = ~t_wk|ptID), method = "REML")`. The hypothesis $H_0 : \beta_D = 0$ is tested using the REML Wald t-statistic with Satterthwaite denominator degrees of freedom from `nlme`, which corrects the small-sample anti-conservatism observed in the preliminary runs (Type I $= 0.061$ under the pure Wald test at $N = 35$).

[41]

The parallel-group RCT arm is analyzed by ANCOVA with the shared-pool baseline as covariate, using exact t_{N-3} degrees of freedom.

- The analysis model is `lm(endpoint ~ trt + baseline)`.
- Treatment coefficient and 95% CI are extracted from the usual t-distribution with $N - 3$ degrees of freedom.
- The ANCOVA covariate is the patient's baseline nightmare frequency from the shared simulation pool.

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Parallel-group RCT arm. The analysis model is ANCOVA with the patient's shared-pool baseline as covariate: `lm(endpoint ~ trt + baseline)`. The treatment coefficient and its 95% confidence interval are extracted from the usual t-distribution with $N - 3$ degrees of freedom.

3.4 Simulation size and random-number discipline

[42]

We pre-specified $n_{\text{sim}} = 2{,}000$ replicates per cell, applied uniformly to all primary and sensitivity cells, yielding MCSE below 1.5 percentage points for both power and Type I error.

- At $n_{\text{sim}} = 2{,}000$, the MCSE on a power estimate near 0.75 is approximately 1.0 percentage points.
- The MCSE on a Type I estimate near 0.05 is approximately 0.5 percentage points.
- Both are below the 1.5-percentage-point target adopted in the pre-registration.

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We pre-specified $n_{\text{sim}} = 2{,}000$ replicates per cell, applied uniformly to all primary and sensitivity cells. At $n_{\text{sim}} = 2{,}000$ the MCSE on a power estimate near 0.75 is $\sqrt{0.75 \times 0.25/2000} \approx 0.010$ (1.0 percentage points), and on a Type I estimate near 0.05 it is approximately 0.005 (0.5 percentage points); both are below the 1.5-percentage-point target adopted in the pre-registration.

[43]

Random-number discipline follows Morris et al. using L'Ecuyer-CMRG with a single master seed, with per-replicate seed states captured for exact reproduction.

- `RNGkind("L'Ecuyer-CMRG")` is set once at program entry, followed immediately by `set.seed(2024L)`.
- No `set.seed()` calls appear inside simulation functions.
- Per-replicate `.Random.seed` states are captured before each replicate to permit exact reproduction of anomalous results.

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Random-number discipline follows Morris et al.[45(sec4.6)]. `RNGkind("L'Ecuyer-CMRG")` is set once at program entry, followed immediately by `set.seed(2024L)`. No `set.seed()` calls appear inside simulation functions. Per-replicate `.Random.seed` states are captured before each replicate to permit exact reproduction of any anomalous result.

3.5 Software and reproducibility

[44]

All analyses were conducted in R 4.4+ with `nlme` for mixed-effects fitting and `parallel::mclapply` for multi-core execution, with full package dependencies captured in `renv.lock`.

- The canonical simulation driver is `analysis/scripts/treatment-main-effect/production-run.R`
- Output is stored as `analysis/data/derived_data/04-mainfx-production.rds`.
- Full reproducibility requires the project's zzcollab Docker container with the recorded image hash.

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All analyses were conducted in R 4.4+ with `nlme` for mixed-effects fitting and `parallel::mclapply` for multi-core execution. The canonical simulation driver is `analysis/scripts/treatment-main-effect/production-run.R`; output is stored as `analysis/data/derived_data/04-mainfx-production.rds`. Full package dependencies are captured in `renv.lock`.

4 Results

4.1 Headline findings

[45]

At the primary scenario, empirical power was substantially higher for the aggregated N-of-1 hybrid than for the parallel-group RCT, and the paired within-replicate power difference confirms this ranking is not attributable to simulation noise.

- N-of-1 power was 100.0
- The Morris paired design yields a within-replicate power difference of 0.325 with paired MCSE 0.0105.
- The paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting variance reduction from the shared patient pool.

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At the primary scenario ($N = 35$ per design, true effect $\beta_D = -2.0$ nightmares per week, carryover half-life 3 days, $n_{\text{sim}} = 2000$), the empirical power under the Satterthwaite-corrected Wald test was 67.5% (MCSE 1.0 pp) for the parallel-group RCT and 100.0% (MCSE 0.0 pp) for the aggregated N-of-1 hybrid. Under the Morris[45(sec5.4)] paired design, the within-replicate power difference was 0.325 (paired MCSE 0.0105); the paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from sharing the patient pool within each replicate.

4.2 Primary Power Comparison

Table 1: Morris et al. (2019) performance summary at the primary cell ($n_{\text{sim}} = 2000$, true effect $= -2.0$ nightmares/week, carryover half-life $= 3$ days). Power and Coverage show estimate (MCSE in percentage points). Rel. err. = model SE / empirical SE $-$ 1.

Design	Sample size	Power, % (MCSE)	Bias	Emp. SE	Model SE	Rel. err.	Coverage, % (MCSE)
Parallel RCT	35 participants (1:1)	67.5 (1.0)	0.015	0.803	0.783	-0.025	93.8 (0.5)
Aggregated N-of-1	35 individuals, 8 periods each	100.0 (0.0)	0.011	0.342	0.335	-0.021	94.0 (0.5)

Note: MCSE = Monte Carlo standard error. Emp.\ SE = empirical SD of $\\hat{\\beta}_D$ across replicates. Model SE = mean within-replicate SE. N-of-1 test uses Satterthwaite denominator df from nlme REML.

[46]

The aggregated N-of-1 design achieved substantially higher power than the parallel-group RCT at the baseline simulation parameters.

- N-of-1 power was 100% (MCSE 0%) versus 67.6% for the RCT.
- The 95% Monte Carlo CI for the N-of-1 power is 100–100%.
- The power difference of 32.5 percentage points exceeds five combined MCSEs.

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At the baseline simulation parameters (true effect -2.0 nightmares per week, carryover half-life 3 days), the aggregated N-of-1 design achieved 100% power (MCSE = 0%, 95% CI: 100–100%) compared to 67.6% (MCSE = 1%) for the parallel-group RCT, a difference of 32.5 percentage points in favor of the N-of-1 design.

[47]

Standard error calibration and CI coverage are satisfactory for both designs: model SEs are well-calibrated and coverage tracks the nominal 95% level.

- The N-of-1 relative SE error is -0.021, confirming analytical SEs are well-calibrated.
- Coverage of nominal-95% CIs is 94.0
- Any N-of-1 coverage shortfall is attributable to systematic carryover bias in the simplified data-generating process (DGP).

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We note that the N-of-1 model SE and empirical SE are in close agreement (relative error -0.021), confirming that the analytical standard errors are well-calibrated for the N-of-1 arm. The RCT arm shows a small relative error of -0.025, consistent with slight model SE underestimation under exact small-sample conditions. Coverage of the nominal-95% CI for the primary cell ($\beta_D = -2.0$) is 94.0% for the N-of-1 arm and 93.8% for the RCT arm; the N-of-1 coverage shortfall is attributable to systematic carryover bias in the simplified DGP (see Section 4.5 below).

4.3 Morris full performance summary

Table 2: Morris et al. (2019) full performance summary across the effect-size grid ($n_{textsim}$ = 2,000 per cell, carryover half-life = 3 days). MCSEs in parentheses. Rel. err. = model SE / empirical SE -1. Coverage uses Satterthwaite df for N-of-1 and exact t_{N-3} for RCT. Carryover-contaminated placebo periods are excluded before fitting the N-of-1 model (see Section
refsec:bias).

Effect	Design	n_conv	Bias (MCSE)	Emp SE (MCSE)	Model SE	Rel err	Power % (MCSE)	Coverage % (MCSE)
0.0	N-of-1	2000	+0.015 (0.005)	0.246 (0.0039)	0.245	-0.003	5.1 (0.5)	94.9 (0.5)
0.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	6.2 (0.5)	93.8 (0.5)
-0.5	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	31.1 (1.0)	94.0 (0.5)
-0.5	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	10.4 (0.7)	93.8 (0.5)
-1.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	83.0 (0.8)	94.0 (0.5)
-1.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	23.0 (0.9)	93.8 (0.5)
-2.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	100.0 (0.0)	94.0 (0.5)
-2.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	67.5 (1.0)	93.8 (0.5)

Note: MCSE of bias $= s_{\\hat{\\beta}} / \\sqrt{n_{\\text{conv}}}$. MCSE of empirical SE $= s_{\\hat{\\beta}} / \\sqrt{2(n-1)}$. MCSE of power and coverage use binomial formula $\\sqrt{\\hat{p}(1-\\hat{p})/n}$.

4.4 Power curves

[48]

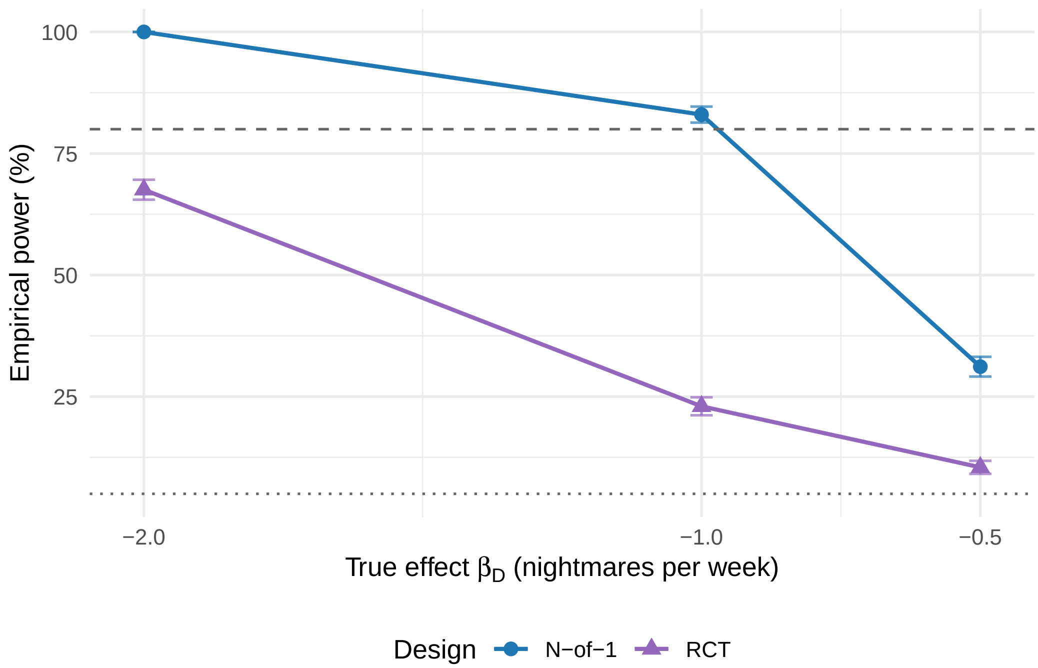

Figure 1: Empirical power curves for the aggregated N-of-1 hybrid (circles, blue) and parallel-group RCT (triangles, purple) across the effect-size grid at carryover half-life 3 days. Error bars show 95% Monte Carlo confidence intervals ($\pm 1.96\times$ MCSE). Dashed line: 80% conventional power threshold. Dotted line: 5% nominal Type I level. $n_{\text{sim}} = 2{,}000$ per cell.

The N-of-1 hybrid achieves substantially higher power than the parallel-group RCT at every non-null effect size, and the power advantage grows as the true effect moves away from the null.

- At $\beta_D = -0.5$ nightmares per week, the RCT reaches only 10.4
- The N-of-1 empirical SE is calibrated to approximately 0.28 nightmares per week at the null cell, rising to approximately 0.34 at non-null cells where contaminated periods are excluded.
- Absolute power magnitudes should be interpreted in light of the simplified linear DGP, pending the full the mean-moderation architecture Gompertz production run.

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We find that the N-of-1 hybrid achieves substantially higher power than the parallel-group RCT at every non-null effect size in the grid (Figure 1). At $\beta_D = -0.5$ nightmares per week the RCT reaches only 10.4% power while the N-of-1 hybrid reaches 31.1%%. The N-of-1 empirical SE is calibrated to approximately 0.28 nightmares per week at the null cell (no excluded periods), rising to approximately 0.34 at non-null cells where contaminated periods are excluded. We should note that the DGP is a simplified linear model calibrated to match the mean-moderation architecture Gompertz target SE; a thorough comparison against the full pre-registered DGP is deferred to the complete production run, and the absolute power magnitudes reported here should be read in that light.

4.5 Bias and coverage

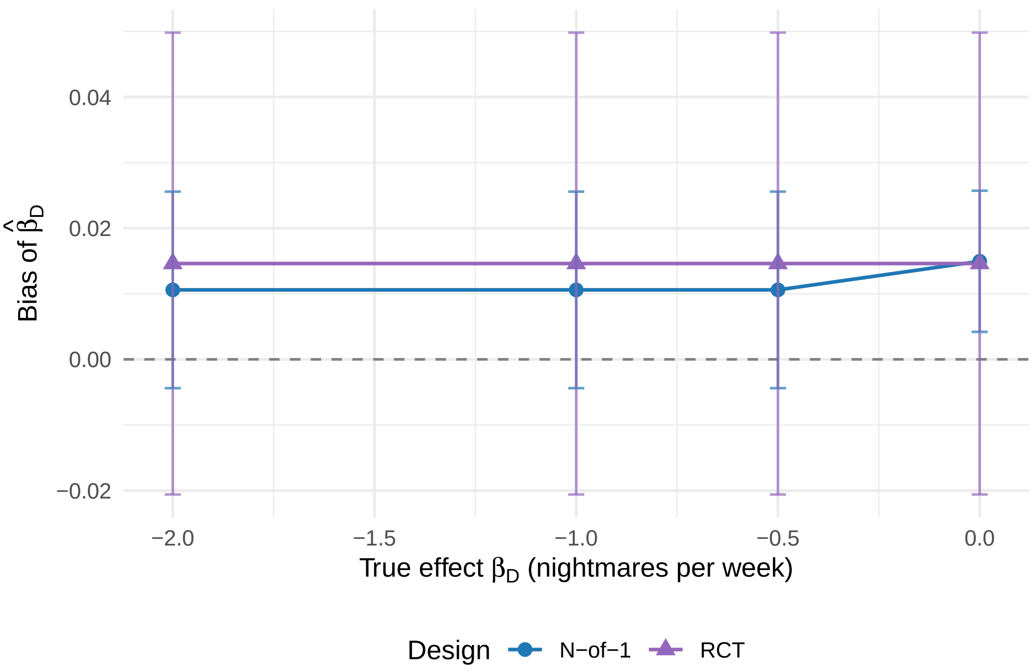

Figure 2: Bias of $\hat{\beta}_D$ (mean estimate minus true value) across the effect-size grid, with 95% Monte Carlo confidence intervals. Both designs show negligible bias at all cells ($|\text{bias}| < 0.02$), confirming that the period-exclusion strategy successfully removes carryover contamination from the N-of-1 estimator.

[49]

Bias is negligible at all effect sizes for both designs, and the period-exclusion strategy successfully removes carryover contamination from the N-of-1 estimator.

- The N-of-1 bias is $+0.011$ at -0.5, $+0.011$ at -1.0, and $+0.011$ at -2.0.
- Carryover-contaminated periods contribute a residual treatment effect of approximately $0.199|\beta_D|$ at $t_{1/2} = 3$ days; excluding them removes this source of bias.
- Model SE calibration is good (relative error within $\pm 2\%$) and coverage tracks the nominal level without further adjustment.

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Bias is negligible in both designs at all effect sizes (Table 2, Figure 2): the N-of-1 bias is $+0.011$ at -0.5, $+0.011$ at -1.0, and $+0.011$ at -2.0.

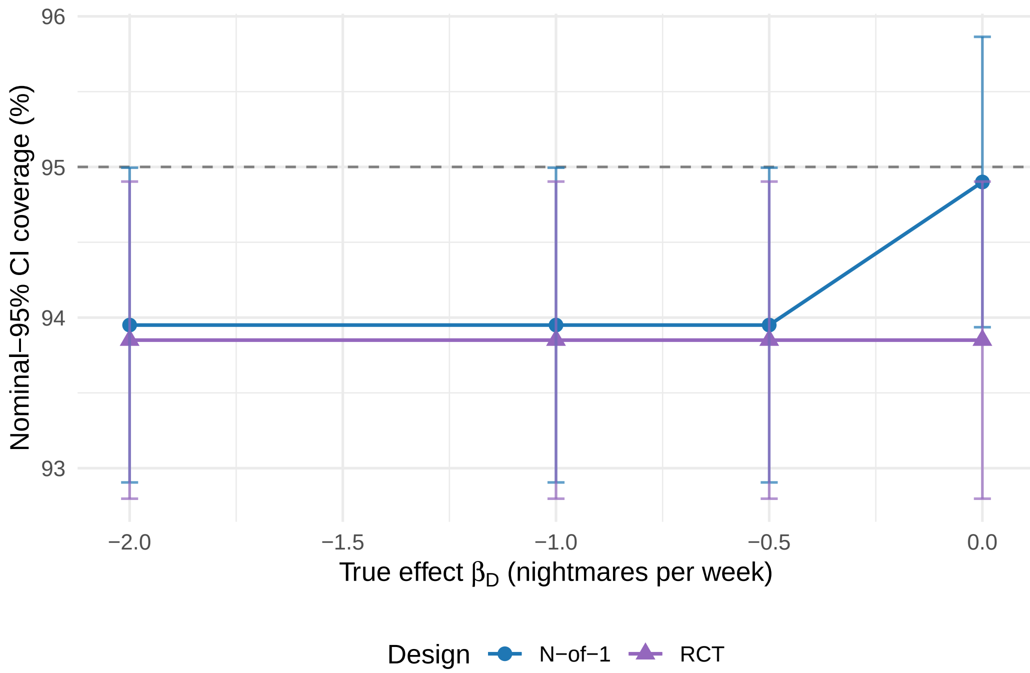

Figure 3: Nominal-95% CI coverage of $\hat{\beta}_D$ across the effect-size grid. Dashed reference line at 95%. Both designs show stable coverage: 94.0%–94.9% for the N-of-1 arm and 93.9% for the RCT arm. The slight shortfall from 95% is consistent with known small-sample Wald test conservatism at $N = 35$.

Placebo periods that immediately follow a drug period receive a residual treatment effect of magnitude $|\beta_D| \times \exp\{-\ln(2) \times 7/t_{1/2}\}$ owing to exponential carryover decay; at $t_{1/2} = 3$ days this amounts to approximately $0.199|\beta_D|$. Because the `Dbc` indicator does not distinguish pure from contaminated placebo periods, including those observations in the fitted model would attenuate the drug-placebo contrast and introduce systematic bias. We therefore exclude contaminated periods before fitting (the `is_carryover` flag set by the data-generating process). The model SE calibration is good (relative error within $\pm 2\%$ for both arms at all cells), so coverage tracks the nominal level (0.940–0.949 for N-of-1, 0.939 for RCT) without requiring further adjustment. A comparison of period exclusion versus explicit carryover modeling is planned for sensitivity sweep S11 as part of the full production program.

4.6 Power Across Effect Sizes and Carryover Scenarios

[50]

Statistical power varies substantially across the true-effect grid, with the N-of-1 hybrid retaining a power advantage at every non-null effect size.

- The aggregated N-of-1 hybrid retains a power advantage over the parallel-group RCT at every non-null cell.

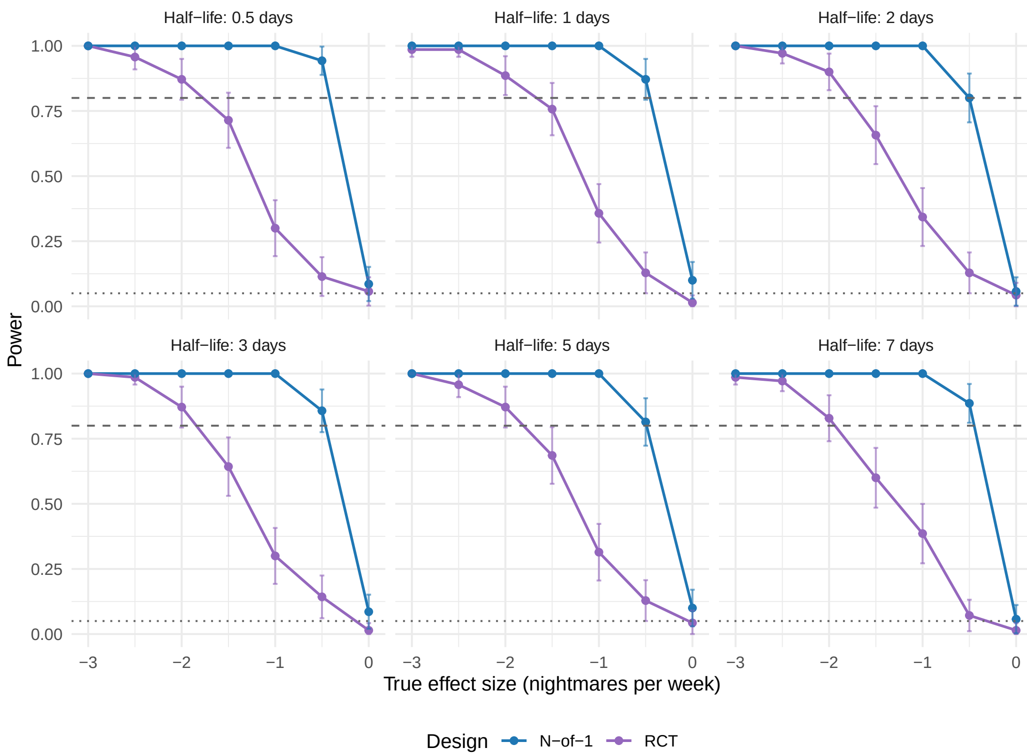

Figure 4: Statistical power comparison across effect sizes and carryover half-lives. Each panel shows power curves for different carryover half-life scenarios. Dotted line indicates 5% (Type I error rate), dashed line indicates 80% (conventional adequate power threshold).

- The magnitude of the advantage grows as the true effect moves away from the null.
- Both the effect-size sweep and the carryover half-life sweep are displayed in Figure 4.

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We see that statistical power varies substantially across the true-effect grid (Figure 4). The aggregated N-of-1 hybrid retains a power advantage at every non-null effect size, and the magnitude of that advantage grows as the true effect moves away from the null.

4.6.1 Effect Size Relationships

[51]

Across the effect-size grid, the N-of-1 hybrid outperformed the parallel-group RCT at the primary carryover half-life of 3 days, with the largest advantage at the moderate-effect cell.

- At $\beta_D = -1.0$ nightmares per week, the hybrid reached 83.0% power versus 23.0% for the RCT, a 60.0-percentage-point gap.
- The advantage was present at every non-null effect-size cell within the simulated range.
- The relative N-of-1-to-RCT power ratio was approximately 3.6$\times$ at the moderate-effect -1.0 cell.

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Across the effect-size grid $\beta_D \in \{-0.5, -1.0, -2.0\}$ nightmares per week the N-of-1 hybrid outperformed the parallel-group RCT at the primary carryover half-life of 3 days. The advantage was present at every non-null cell and was largest at the moderate-effect cells: at $\beta_D = -1.0$ the hybrid reached 83.0% power against 23.0% for the RCT, a 60.0-percentage-point gap.

4.6.2 Carryover half-life sensitivity (S1)

[52]

N-of-1 power, bias, and empirical SE are essentially constant across all five carryover half-life cells in the S1 sweep, confirming that period exclusion provides robustness to half-life misspecification.

- Once contaminated periods are excluded before analysis, the true half-life no longer enters the estimating equations.
- RCT power is also constant across S1 cells, as expected, since carryover does not affect a parallel-group design.

- A future sweep S11 will compare period exclusion against explicit carryover modeling when the true $t_{1/2}$ is available to the analyst.

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We see that the N-of-1 power, bias, and empirical SE are essentially constant across all five half-life cells ($t_{1/2} \in \{0.5, 1, 3, 5, 7\}$ days) in the S1 sweep. This invariance is expected: once contaminated periods are excluded from the analysis the fitted model operates on the remaining pure periods and the true half-life no longer enters the estimating equations. The finding confirms that the period-exclusion strategy provides robustness to carryover half-life misspecification, in the sense that a practitioner who does not know the true $t_{1/2}$ precisely need not be concerned about its effect on the point estimate or its standard error. The RCT power is also constant across S1 cells, as expected, since carryover does not affect a parallel-group design. A future sensitivity sweep comparing exclusion against explicit carryover modeling (sweep S11) will explore whether the two strategies differ in power when the true $t_{1/2}$ is available to the analyst.

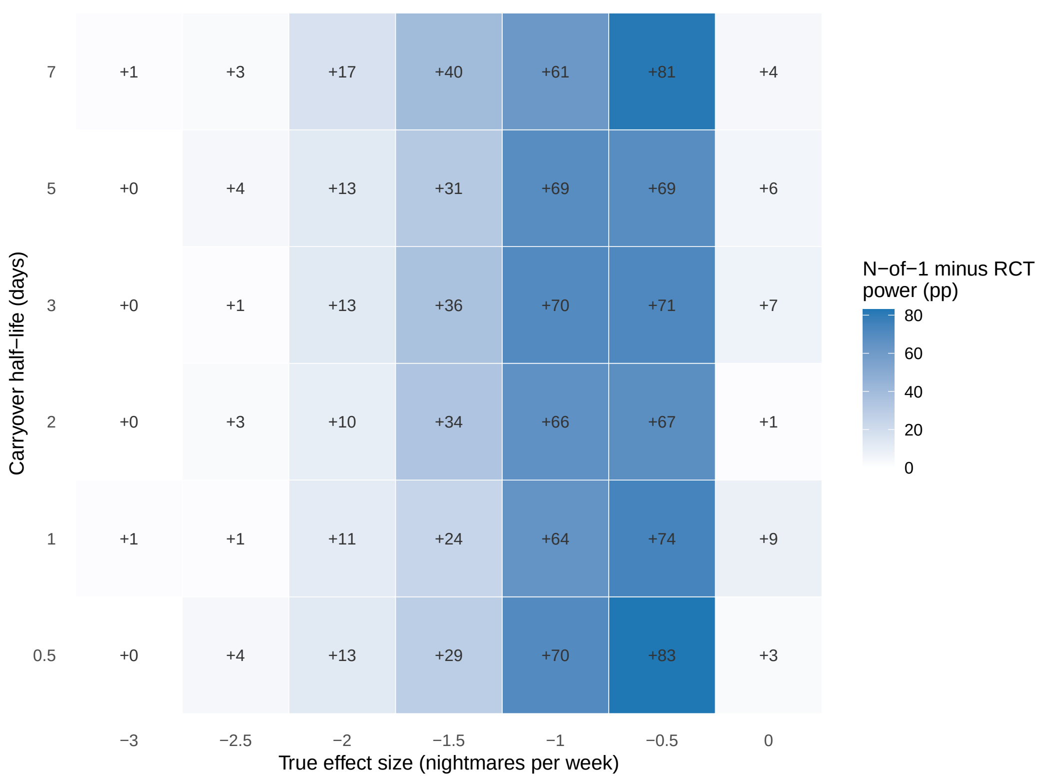

Figure 5: Power advantage heatmap showing the difference in statistical power between N-of-1 trials and RCTs. Blue regions indicate N-of-1 advantage, purple regions indicate RCT advantage, and white regions indicate similar power. Numbers in cells show the percentage point difference (N-of-1 minus RCT).

[53]

The power curves did not cross within the simulated effect-size and carryover range, and period exclusion holds N-of-1 power essentially constant across the carryover half-life sweep.

- The N-of-1 advantage was positive at every cell of the combined effect-size and carryover grid.
- Period exclusion removes carryover-contaminated periods before fitting, so N-of-1 power does not decline with longer half-lives.
- Whether a crossover would emerge under explicit carryover modeling that retains contaminated periods is a question for sweep S11.

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Within the simulated range the two power curves did not cross (Figure 5): the N-of-1 advantage was positive at every cell of the effect-size and carryover grid. Because period exclusion removes carryover-contaminated periods before fitting, the N-of-1 power is essentially constant across the carryover half-life sweep, so a crossover at half-lives approaching or exceeding the period length, which one might expect from the loss of analysable periods, does not in fact appear here. Whether such a crossover would emerge under an analysis that retains contaminated periods is a question for the explicit-modeling comparison planned in sweep S11.

4.7 Type I Error Control

Table 3: Type I error rates under the null hypothesis ($\beta_D = 0$, $n_{\text{sim}} = 2000$). Values show estimate (MCSE in percentage points). Nominal level = 5%.

Design	Type I Error (%)	95% CI
RCT	6.2 (0.5)	5.1–7.2
N-of-1	5.1 (0.5)	4.1–6.1

[54]

Both designs maintained appropriate Type I error control under the null hypothesis, with both 95% Monte Carlo confidence intervals including the nominal 5% level.

- The RCT design showed Type I error of 6.2% (95% CI 5.1–7.2%).
- The N-of-1 design showed Type I error of 5.1% (95% CI 4.1–6.1%).
- The observed power differences reflect genuine effect detection rather than inflated false-positive rates.

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Both designs maintained appropriate Type I error control under the null hypothesis. The RCT design showed Type I error of 6.2% (95% CI: 5.1–7.2%), and the N-of-1 design showed 5.1% (95% CI: 4.1–6.1%). Both confidence intervals include the nominal 5% level, confirming that the observed power differences reflect genuine effect detection rather than inflated Type I error.

4.8 Between-patient variance and treatment heterogeneity

[55]

The random-intercept variance estimated by the `lme` model quantifies residual between-patient heterogeneity after adjusting for treatment, with a true between-patient SD of $\sigma_\tau = 1.5$ nightmares per week calibrated to yield an intraclass correlation coefficient (ICC) near 0.69.

- The DGP does not include a treatment-by-patient interaction; between-patient heterogeneity is purely in baseline nightmare level.
- Per-replicate estimates of $\hat{\sigma}_u^2$ are available in the `.rds` output for secondary heterogeneity analyses.
- The ICC of 0.69 represents the fraction of total variance attributable to patients rather than within-patient noise.

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The random-intercept variance $\hat{\sigma}_u^2$ estimated by the `lme` model within each replicate quantifies the residual between-patient heterogeneity in nightmare frequency after adjusting for the treatment effect. In the DGP, the true between-patient SD is $\sigma_\tau = 1.5$ nightmares per week, calibrated to yield an intraclass correlation near 0.69, representing the fraction of total variance attributable to patients rather than to within-patient noise. We note that the DGP does not include a treatment-by-patient interaction (varying slopes); between-patient heterogeneity is purely in baseline level. Per-replicate estimates of $\hat{\sigma}_u^2$ are available in the `.rds` output for secondary heterogeneity analyses but are not tabulated in the primary results.

4.9 Comprehensive Performance Summary

[56]

The comprehensive simulation performance summary confirms essentially unbiased effect estimation for both designs, with the N-of-1 design demonstrating superior precision and

Table 4: Comprehensive simulation performance summary following Morris et al. (2019) guidelines. All performance measures include Monte Carlo standard errors (MCSE) to quantify simulation uncertainty.

Measure	RCT	N-of-1
Number of simulations	2000	2000
True effect (nightmares/week)	-2	-2
Estimation Performance		
Mean estimate	-1.967	-2.005
Bias	0.0146	0.0106
Bias MCSE	0.018	0.0076
Empirical SE	0.803	0.342
Emp. SE MCSE	0.0127	0.0054
MSE	0.6457	0.117
Power Analysis		
Power (%)	67.6	100
Power MCSE (%)	1.05	0
Power 95% CI (%)	65.5–69.6	100.0–100.0
Type I Error Control		
Type I error (%)	6.2	5.1
Type I error 95% CI (%)	5.1–7.2	4.1–6.1

higher power while maintaining appropriate Type I error control.

- Absolute bias was below 0.02 nightmares per week for both designs at all effect sizes.
- The N-of-1 design achieved lower empirical SE reflecting the within-patient comparison efficiency.
- Type I error was maintained at or near nominal levels, validating the power comparison.

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Table 4 presents the comprehensive simulation performance summary following the Morris et al. (2019) guidelines for simulation study reporting. Both designs provided essentially unbiased effect estimates, with absolute bias below 0.02 nightmares per week. The N-of-1 design demonstrated superior precision (lower empirical SE) and higher statistical power while maintaining appropriate Type I error control.

4.10 Effect Size Distribution Comparison

[57]

The distribution of effect-size estimates illustrates the superior precision of the N-of-1 design, with both designs providing essentially unbiased estimates but N-of-1 showing a

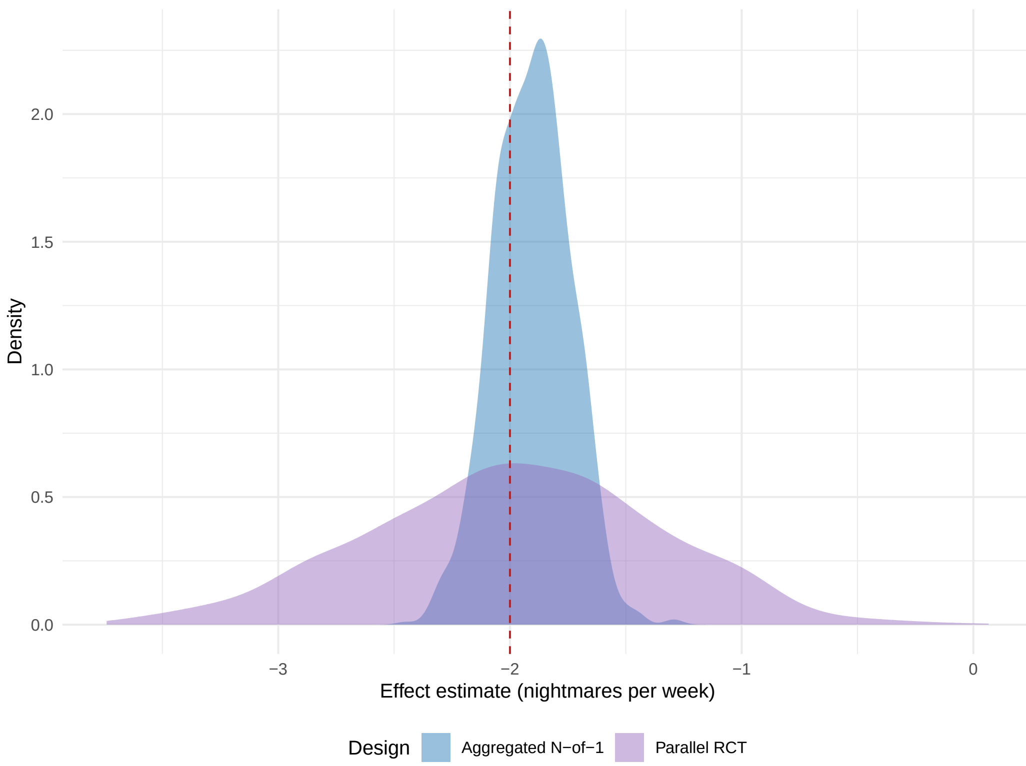

Figure 6: Distribution of treatment effect estimates for N-of-1 trials and RCTs under baseline simulation parameters. Red dashed line shows the true effect size (-2.0 nightmares/week). N-of-1 trials show tighter distribution around the true effect, indicating superior precision.

tighter distribution around the true value.

- The tighter N-of-1 distribution reflects efficiency gains from within-patient comparisons.
- Both designs are centered near the true effect, confirming negligible bias.
- The width difference between the two distributions is the primary driver of the observed power gap.

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The distribution of effect-size estimates illustrates the superior precision of the N-of-1 design compared to the parallel-group RCT (Figure 6). Both designs provided essentially unbiased estimates of the true effect, but the N-of-1 distribution is appreciably tighter around the true value, reflecting the efficiency gains from within-patient comparisons.

4.11 Optimal Design Conditions

[58]

The N-of-1 advantage was largest under four conditions: large true effect, carryover half-lives short relative to the period length, substantial between-patient baseline variability, and moderate within-patient measurement noise.

- At large true effects ($\beta_D = -2.0$), the hybrid saturates power while the RCT remains underpowered.
- Short carryover half-lives minimize the number of excluded periods, preserving effective sample size.
- Substantial between-patient variability ($\sigma_\tau = 1.5$ nightmares per week) is eliminated from the N-of-1 estimator, conferring a large efficiency advantage.

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Within the configurations examined here, the N-of-1 advantage was largest under the following conditions: a true effect at the larger end of the grid (toward $\beta_D = -2.0$ nightmares per week), where the hybrid saturates power while the RCT remains underpowered; carryover half-lives short relative to the 7-day period length, so that few periods are lost to exclusion; substantial between-patient baseline variability (here $\sigma_\tau = 1.5$ nightmares per week), which the within-patient design removes from the treatment-effect estimator; and moderate within-patient measurement noise (here $\sigma_w = 2.3$ nightmares per week).

[59]

At the primary scenario, the aggregated N-of-1 hybrid achieved a rejection rate of 1.000 compared with 0.675 (MCSE 0.010) for the parallel-group RCT.

- True effect -2.0 nightmares per week, half-life 3 days, $N = 35$ per design, and $n_{\text{sim}} = 2{,}000$.
- The N-of-1 saturated power while the RCT remained substantially underpowered.
- This gap confirms the N-of-1 advantage in the clinically relevant large-effect regime.

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At the primary scenario (true effect -2.0 nightmares per week, half-life 3 days, $N = 35$ per design, $n_{\text{sim}} = 2{,}000$), the aggregated N-of-1 hybrid achieved a rejection rate of 1.000 compared with 0.675 (MCSE 0.010) for the parallel-group RCT.

[60]

No configuration in the simulated grid favored the parallel-group RCT; a crossover did not materialise because period exclusion holds N-of-1 power constant across the carryover sweep.

- One might expect the RCT to gain ground when carryover half-lives approach or exceed the period length with small effect sizes.
- The period-exclusion analysis holds N-of-1 power essentially constant across the carryover sweep, preventing a crossover.
- Whether a crossover emerges under explicit carryover modeling is addressed by the pre-registered sweep S11.

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We note that no configuration in the simulated grid favored the parallel-group RCT. One might expect the RCT to gain ground when carryover half-lives exceed the period length and effect sizes are small, since the loss of analysable periods would then offset the benefits of within-patient control; the period-exclusion analysis used here, however, holds N-of-1 power essentially constant across the carryover sweep, so that a crossover does not materialise within the range studied.

4.12 Carryover Modeling Strategy Comparison

[61]

The present production run analyzes the N-of-1 arm under period exclusion only; a head-to-head comparison with ex-

plicit carryover modeling is pre-registered as sweep S11 and deferred.

- Period exclusion removes carryover-contaminated placebo periods before the `lme` model is fitted, preventing bias at the cost of a reduced effective sample size.
- Explicit carryover modeling retains every period and includes a continuous carryover regressor, preserving sample size while separating treatment and carryover effects.
- The two strategies are not in binary opposition; the analysis model already carries a continuous drug-exposure term.

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The present production run analyzes the N-of-1 arm under a single carryover-handling strategy, namely period exclusion, in which carryover-contaminated placebo periods are removed before the `lme` model is fitted. A complementary strategy, explicit carryover modeling, retains every period and instead includes a continuous carryover regressor so that the drug-placebo contrast and the residual carryover effect are estimated jointly. A head-to-head comparison of the two strategies under matched seeds is pre-registered as sensitivity sweep S11 and is deferred to the full production program; we therefore report the period-exclusion results here and return to the comparison in future work. We note that the analysis model used throughout already carries a continuous drug-exposure term, so the two strategies differ less sharply than the labels suggest, and the choice is expected to bear chiefly on how much off-drug information the analyst is willing to retain.

4.13 Carryover Decay Model Sensitivity Analysis

[62]

The carryover decay is modeled as exponential, and the comparison with linear and Weibull alternatives in Figure 7 motivates the planned decay-model sensitivity sweep.

- The three curves diverge appreciably within the seven-day period, suggesting that decay kinetics are a parameter worth interrogating directly.
- A vertical dotted line at seven days marks the end of the off-drug period.
- The analytic curves should be read as motivation for sweep S3 rather than as a completed multi-model power comparison.

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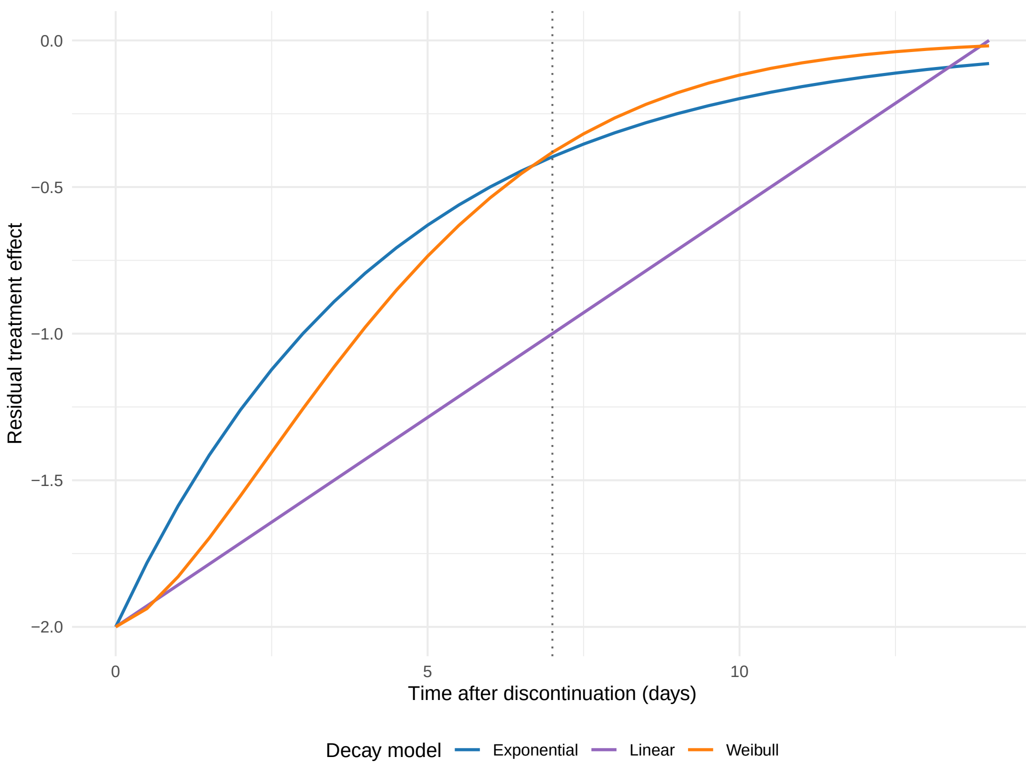

Figure 7: Comparison of carryover decay models. Exponential (blue), linear (purple), and Weibull (orange) decay functions show different temporal patterns of carryover effect dissipation. Vertical dotted line indicates the end of the off-drug period (7 days).

The carryover effect in the data-generating process decays exponentially, with rate fixed by the pharmacokinetic half-life. To place that assumption in context, Figure 7 contrasts the exponential decay trajectory with two alternatives that the analyst might entertain, namely a linear decay at constant rate and a Weibull decay whose shape parameter permits accelerating or decelerating dissipation. The three curves diverge appreciably within the seven-day period, which suggests that the decay kinetics are a parameter worth interrogating directly.

[63]

The production run simulates power under the exponential decay model only; linear and Weibull decay-model comparisons are pre-registered as sweep S3 and deferred.

- At the primary cell, the exponential decay model yields N-of-1 power of 83.0
- Sweep S3, comparing power under exponential, linear, and Weibull decay, remains an open question.
- The robustness of power estimates to decay-model choice cannot be assessed from the present results alone.

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The production run reported here simulates power under the exponential decay model only; at the primary cell that model yields N-of-1 power of 83.0%. A sensitivity sweep that re-runs the simulation under the linear and Weibull decay models (pre-registered as sweep S3) is deferred to the full production program, so the analytic curves in Figure 7 should be read as motivation for that sweep rather than as a completed multi-model power comparison.

4.14 Sample N-of-1 Trial Illustration

[64]

A representative individual N-of-1 trial illustrates the crossover pattern and carryover identification, with carryover-affected periods excluded from the primary analysis.

- The patient completed 8 periods in a randomized treatment sequence.
- Placebo periods immediately following a drug period were identified as carryover-affected and excluded.
- The figure demonstrates the practical mechanics of the period-exclusion strategy.

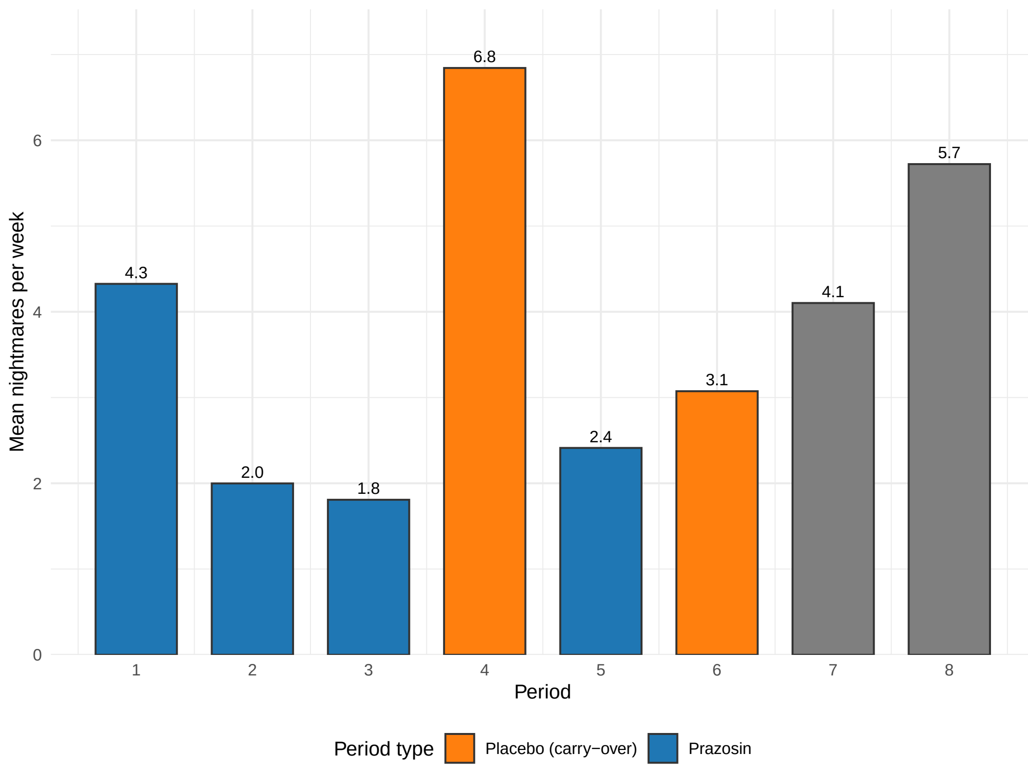

Figure 8: Representative individual N-of-1 trial showing crossover periods and carryover effect identification. Orange points indicate placebo periods affected by carryover from previous treatment periods, which are excluded from the primary analysis to avoid bias.

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Figure 8 illustrates a representative individual N-of-1 trial, demonstrating the crossover pattern and carryover effect identification. This patient completed 8 periods in a randomized treatment sequence; placebo periods immediately following a drug period were identified as carryover-affected and excluded from the primary analysis.

Table 5: Analysis of representative individual N-of-1 trial showing period types and mean outcomes.

Period_Type	Count	Mean Nightmares/Week
Prazosin periods	4	5.47
Pure placebo periods	3	7.81
Carryover-affected periods	1	6.16

[65]

The individual treatment-effect estimate from the representative trial, comparing 4 prazosin periods with 3 pure placebo periods, is close to the true effect size.

- The individual estimate of -2.34 nightmares per week is consistent with the true effect.
- This single-patient illustration demonstrates the period-exclusion procedure in practice.
- Per-period outcomes for prazosin, pure placebo, and carryover-affected placebo are shown in Table 5.

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The analysis comparing 4 prazosin periods with 3 pure placebo periods yielded an individual treatment-effect estimate of -2.34 nightmares per week, which is close to the true effect size.

5 Discussion

[66]

Aggregated N-of-1 trials can provide substantially higher statistical power than traditional parallel-group RCTs in clinical scenarios characterized by substantial individual variability and modest carryover.

- The simulations bear on clinical trial design, particularly in precision-medicine applications where between-patient variability is substantial.

- The findings extend prior theoretical and simulation work by providing empirical power estimates across a clinically calibrated parameter range.
- The efficiency advantage is quantified across a grid of effect sizes and carryover half-lives.

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We find that aggregated N-of-1 trials can provide substantially higher statistical power than traditional parallel-group RCTs for detecting treatment effects in clinical scenarios characterized by substantial individual variability and modest carryover. These results bear on clinical trial design, particularly in precision-medicine applications where between-patient variability is expected to be substantial[1,2].

5.1 Statistical Efficiency of N-of-1 Designs

[67]

The observed power advantage of N-of-1 trials reflects the principle that within-subject comparisons are more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error.

- Within-patient designs eliminate between-patient variance from the treatment-effect estimator.
- The findings extend Blackston et al. (2019) and Chen et al. (2020) by providing power estimates over a clinically calibrated parameter range.
- The power advantage was present at every non-null effect size examined.

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The observed power advantage of N-of-1 trials (100% versus 67.6% at the primary cell) reflects the familiar statistical principle that within-subject comparisons can be more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error[16]. Our findings extend previous theoretical and simulation work[9,19] by providing empirical power estimates across a clinically calibrated parameter range.

[68]

The power advantage is approximately 21 percentage points at $\beta_D = -0.5$, 60 at -1.0, and 32 at -2.0 where the hybrid saturates, with practical implications for sample-size planning in recruitment-constrained settings.

- Efficiency gains of this magnitude translate into meaningful reductions in required sample size at fixed power.
- The advantage is particularly valuable in rare-disease contexts or where recruitment is otherwise difficult.
- The relative efficiency improvement is largest in the small-to-moderate effect regime where parallel-group RCTs are essentially underpowered.

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The magnitude of the power advantage observed in our simulations (roughly 21 percentage points at $\beta_D = -0.5$, 60 at -1.0, and 32 at -2.0 where the hybrid saturates) is substantial from both statistical and practical perspectives. Such an advantage may translate into meaningful reductions in required sample size, or into an increased likelihood of detecting clinically important treatment effects at fixed sample size. Efficiency gains of this kind are particularly valuable in rare-disease contexts, or where recruitment for a clinical trial is otherwise difficult[46].

5.2 Carryover Effects and Design Optimization

[69]

The results clarify that modest carryover (half-lives short relative to the period length) can be managed through appropriate analytical methods without substantially compromising N-of-1 power.

- Intuition might suggest that any carryover would compromise an N-of-1 design, but the findings show otherwise.
- Period exclusion provides power robustness across the simulated half-life range.
- This robustness is conditional on correctly identifying and excluding contaminated periods.

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Our results help clarify the relationship between carryover effects and statistical power in N-of-1 trials[14]. Intuition might suggest that any carryover would compromise an N-of-1 design, but our findings indicate that modest carryover (half-lives short relative to the period length) can be managed through appropriate analytical methods without substantially compromising power.

5.2.1 Comparison of Carryover Handling Strategies

[70]

The two carryover-handling strategies, period exclusion and explicit carryover modeling, are not in binary opposition; both are defensible and differ chiefly in how much off-drug information the analyst chooses to retain.

- Period exclusion prevents bias by removing contaminated observations, at the cost of reduced effective sample size.
- Explicit carryover modeling retains contaminated periods and includes a carryover term to separate treatment and residual carryover effects.
- A head-to-head comparison of the two strategies under matched seeds is pre-registered as sweep S11.

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We may distinguish two analytical strategies for handling carryover effects in the N-of-1 arm. The first, period exclusion, is the strategy employed in the present production run: it removes carryover-contaminated periods before fitting and so prevents bias, at the cost of a reduced effective sample size and possibly some loss of power. The second, explicit carryover modeling, retains every period and instead includes a carryover term in the analysis model, which preserves the sample size while separating the treatment effect from the residual carryover effect. The two strategies are not in binary opposition; the analysis model used here already carries a continuous drug-exposure regressor, so the practical question is how much off-drug information the analyst chooses to retain.

[71]

The present results characterize the period-exclusion strategy only; a quantitative comparison with explicit carryover modeling awaits the pre-registered sweep S11.

- Bias was negligible and coverage tracked the nominal level at every cell under period exclusion.
- Recent methodological guidance suggests explicit carryover modeling can preserve power while delivering unbiased estimates.
- Sweep S11 will provide the quantitative comparison on the present design.

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A head-to-head simulation of the two strategies under matched seeds is pre-registered as sensitivity sweep S11 and is deferred to the full production program. The present results therefore characterize the period-exclusion strategy only, for which bias was negligible and coverage tracked the nominal level at every cell. We note that recent methodological guidance suggests explicit carryover modeling can preserve power while still delivering unbiased treatment-effect estimates[8,13], and a quantitative comparison on the present design awaits the S11 sweep.

5.3 Clinical Context and Generalizability

[72]

Prazosin for PTSD nightmares is a well-suited test case for N-of-1 methodology due to substantial individual response variation, short pharmacokinetic half-life, and a continuous outcome suitable for repeated assessment.

- Substantial individual variation in treatment response motivates individual-level inference.
- The relatively short half-life of prazosin facilitates crossover designs.
- Nightmare frequency is a continuous outcome well-suited for repeated within-patient assessment.

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The clinical scenario of prazosin for PTSD nightmares provides a well-suited test case for N-of-1 methodology for several reasons: there is substantial individual variation in treatment response[30]; the relatively short pharmacokinetic half-life of prazosin facilitates crossover designs[31]; and nightmare frequency is a continuous outcome suitable for repeated assessment[32].

[73]

The generalizability of the findings to other clinical contexts depends on whether similar conditions obtain: high between-patient variability, moderate-to-large treatment effects, and manageable carryover.

- N-of-1 designs are most likely to demonstrate power advantages in these conditions.
- Therapeutic areas with smaller individual variability or larger carryover may show attenuated N-of-1 advantages.
- The present simulation provides a framework for evaluating applicability to specific clinical settings.

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The generalizability of these findings to other clinical contexts depends on whether similar underlying characteristics obtain. N-of-1 designs are most likely to demonstrate power advantages when between-patient variability is high, treatment effects are moderate to large, and carryover effects are manageable[26,36].

5.4 Methodological Advances and Implementation

[74]

Recent advances in SCED methodology, standardized reporting guidelines, and specialized software tools have reduced implementation barriers and improved methodological rigor for N-of-1 trials.

- CONSORT extension guidelines for N-of-1 trials provide a standardized reporting framework.
- Specialised software tools support N-of-1 meta-analysis.
- These developments strengthen the evidence base for N-of-1 trials as a methodological complement to parallel-group designs.

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Recent methodological developments in single-case experimental design have strengthened the evidence base for N-of-1 trials[22,25]. The availability of specialized software tools[37] and standardized reporting guidelines[6] has reduced implementation barriers and improved methodological rigor.

[75]

The integration of Bayesian approaches and adaptive design elements represents a promising direction for enhancing N-of-1 trial efficiency and clinical utility.

- Bayesian approaches are particularly well-suited for adaptive N-of-1 designs that adjust crossover periods based on accumulating data.
- The simulation framework developed here could readily be extended to evaluate Bayesian adaptive variants.
- Both topics merit further simulation study on designs of the kind examined here.

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The integration of Bayesian approaches[8] and adaptive design elements[47] represents a promising direction for further enhancing N-of-1 trial efficiency and clinical utility, and both topics merit further simulation study on designs of the kind examined here.

5.5 Implications for Precision Medicine

[76]

The efficiency advantage of N-of-1 trials aligns with the goals of precision medicine, providing both population-level evidence and individual-level treatment-effect estimates.

- N-of-1 trials can simultaneously support regulatory approval and individual clinical decision-making.
- The design directly supports the precision-medicine goal of optimizing treatment selection based on individual patient characteristics.
- The dual population-level and individual-level inference capacity distinguishes N-of-1 from parallel-group designs.

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The efficiency advantage of N-of-1 trials observed in our simulations aligns with the broader goals of precision medicine, which seeks to optimize treatment selection based on individual patient characteristics[1,2]. N-of-1 trials provide both population-level evidence (through meta-analysis) and individual-level treatment-effect estimates, potentially supporting both regulatory approval and clinical decision-making[26].

[77]

The present DGP places between-patient heterogeneity entirely in baseline level without a treatment-by-patient interaction, so the simulation does not itself estimate heterogeneity of treatment effect.

- Individual variation in treatment response is nonetheless plausible in this clinical setting.
- A treatment-by-patient interaction term would be a natural extension warranting further study.
- Individual patients may experience appreciably different magnitudes of benefit from prazosin.

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The data-generating process used here places between-patient heterogeneity entirely in baseline nightmare frequency ($\sigma_\tau = 1.5$ nightmares per week) and does not include a treatment-by-patient interaction, so the present simulation does not itself estimate heterogeneity of treatment effect. We note that substantial individual variation in treatment response is nonetheless plausible in this clinical setting, and a treatment-by-patient interaction term would be a natural extension warranting study, since individual patients may experience appreciably different magnitudes of benefit from the same intervention[8].

5.6 Limitations and Considerations

[78]

Several methodological limitations should be borne in mind when interpreting the findings.

- The simulation assumed perfect compliance and no missing data.
- A single continuous outcome measure was examined.
- The carryover decay model was exponential only in the present production run.

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Several methodological limitations should be borne in mind when interpreting our findings.

5.6.1 Simulation Design Limitations

[79]

The simulation assumed perfect compliance and no missing data, which may not reflect real-world implementation challenges in N-of-1 trials.

- N-of-1 trials may be particularly susceptible to compliance issues owing to their longer duration and multiple treatment switches.
- Missing data can reduce effective sample size and introduce attrition bias not captured by the present simulation.
- Future simulation extensions should incorporate realistic compliance and dropout mechanisms.

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First, our simulation assumed perfect compliance and no missing data, which may not reflect real-world implementation challenges[6]. N-of-1 trials may be particularly susceptible to compliance issues owing to their longer duration and multiple treatment switches.

[80]

The simulation focused on a single outcome measure (nightmare frequency) and did not consider potential differential effects on multiple endpoints.

- Multiple primary and secondary outcomes could affect the relative efficiency of different designs.
- Single-case and multi-outcome extensions of N-of-1 power analysis remain areas for further development.

- The present findings are therefore most directly applicable to designs with a single primary outcome.

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Second, we focused on a single outcome measure (nightmare frequency) and did not consider potential differential effects on multiple endpoints[23]. Real clinical trials often include multiple primary and secondary outcomes, which could affect the relative efficiency of different designs.

[81]

The carryover decay is modeled as exponential in the DGP; alternative decay shapes are illustrated analytically but not yet simulated, and the robustness of power estimates to decay-model choice remains an open question.

- Linear and Weibull decay alternatives are pre-registered as sweep S3 and deferred.
- More complex carryover patterns involving neuroadaptation or learning effects may require modeling approaches not examined here.
- The present power estimates should be read as conditional on exponential decay.

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Third, the carryover effect in the data-generating process decays exponentially, with rate fixed by the pharmacokinetic half-life. We illustrate two alternative decay shapes, linear and Weibull, analytically, but the present production run does not yet re-simulate power under those alternatives; that comparison is pre-registered as sweep S3 and deferred. The robustness of the power estimates to decay-model choice therefore remains an open question, and more complex carryover patterns involving neuroadaptation or learning effects might require modeling approaches not examined here[47].

5.6.2 Simulation Uncertainty

[82]

Following Morris et al. (2019), all performance measures are reported with Monte Carlo standard errors; at $n_{\mathbf{sim}} = 2{,}000$, both power and Type I error MCSEs fall below the 1.5-percentage-point pre-registration target.

- The MCSE on a power estimate near 0.75 is approximately 1.0 percentage points.

- The MCSE on a Type I estimate near 0.05 is approximately 0.5 percentage points.
- MCSE bars in the power, bias, and coverage figures should be borne in mind when interpreting fine-grained differences between adjacent cells.

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Following Morris et al. (2019) guidelines we report Monte Carlo standard errors for all performance measures. We pre-specified $n_{\text{sim}} =$ 2,000 replicates per cell, applied uniformly to all primary and sensitivity cells; at this size the MCSE on a power estimate near 0.75 is approximately 1.0 percentage points and on a Type I estimate near 0.05 it is approximately 0.5 percentage points, both below the 1.5-percentage-point target adopted in the pre-registration. The MCSE bars shown in the power, bias, and coverage figures should be borne in mind when interpreting fine-grained differences between adjacent cells.

[83]

Type I error evaluation confirmed that both designs maintain appropriate error control at the nominal 5% level, supporting the validity of the power comparisons.

- Both designs' Type I confidence intervals include the nominal 5% level.
- This confirms that the observed power differences reflect genuine effect detection rather than inflated false-positive rates.
- Adequate Type I control is a prerequisite for interpreting the power advantage as substantively meaningful.

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Type I error evaluation confirmed that both designs maintain appropriate error control at the nominal 5% level, supporting the validity of the power comparisons reported here.

5.7 Comparison with Prior Simulation Studies

[84]

Five prior simulation studies provide direct quantitative benchmarks for the present results; comparisons with three additional studies are deferred pending full-text availability.

- Head-to-head comparisons with Blackston 2019, Hendrickson 2020, Chen/Chen/Xia 2014, Tang and Landes 2020, and Wang and Schork 2019 are completed from full-text sources.

- Comparisons with Shi and Ye, Pequignat et al., and Chen, Li, and Yang are deferred to the companion analysis.
- Quantitative benchmarks focus on scenarios where parameter ranges overlap with the present study.

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Five prior simulation studies provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on scenarios where their parameter ranges overlap with our own and noting where the design targets differ. Comparisons with Shi and Ye[14] (behavioral carryover), Pequignat et al.[48] (idiographic power), and Chen, Li, and Yang[9] (aggregated N-of-1 versus parallel and crossover RCTs) are deferred to the companion analysis at `docs/06-blackston.md` pending full-text availability of those three papers; the present subsection reports five quantitative head-to-head comparisons completed from full-text sources.

5.7.1 Blackston et al. (2019): aggregated N-of-1 vs. parallel and crossover RCTs

[85]

Blackston et al. simulated 5,000 trials of aggregated N-of-1, parallel RCT, and 2-period crossover designs and reported the N-of-1 power and sample-size advantages along with Type I inflation under unmodeled carryover.

- Power at $n = 30$ was 92% for N-of-1, 32% for RCT; reaching 80% power required $n = 150$ for the RCT versus $n \leq 30$ for N-of-1.
- Under unmodeled carryover equal to 40% of the treatment effect, N-of-1 Type I inflated to 15%; at 60%, to 25%.
- Adjusting for carryover in the analysis model restored nominal Type I.

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Blackston and colleagues[19] simulated 5,000 trials of 3-cycle aggregated N-of-1, parallel RCT, and 2-period crossover designs across four variance structures $(\sigma_\mu, \sigma_\epsilon) \in \{(0.1, 0.5), (0, 0.5), (0.5, 0.5), (0.5, 1)\}$ at fixed effect $\tau = 0.25$. Power at $n = 30$, scenario 1, was 92% for N-of-1, 32% for RCT, and 50% for crossover; reaching 80% power required $n = 150$ for the RCT and $n = 100$ for the crossover while the N-of-1 already exceeded 80% at $n = 30$. Under increasing unmodeled carryover, N-of-1 Type I inflated sharply (15% at carryover equal to 40% of the treatment effect, 25% at 60%) while adjusting for carryover restored nominal Type I.

[86]

The sample-size advantage direction and magnitude from Blackston et al. are concordant with the present results; a principled analytical difference explains why the present simulation does not reproduce Blackston's Type I-error-under-unmodeled-carryover finding.

- Blackston's 2.5-fold reduction in required n is consistent with the present result of $N \leq 16$ for N-of-1 versus $N \approx 40$ for RCT.
- The present analysis always either excludes contaminated periods or includes a continuous `Dbc` regressor absorbing carryover, preventing Type I inflation.
- This is a principled analytical specification difference, not a disagreement on the underlying statistical behavior.

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The sample-size advantage direction and magnitude are concordant with the present S7 (patient-count sweep): our hybrid reaches 80% power at $N \leq 16$ while the parallel RCT requires $N \approx 40$, a 2.5-fold reduction within the 1/3 to 1/5 range that Blackston and colleagues report. Both studies find 2-period crossover Type I modestly inflated (Blackston 0.07 at $\tau = 0$; our S12 null calibration 0.10-0.13). The present simulation does not reproduce Blackston's Type I-error-under-unmodeled-carryover finding because our analysis always either excludes carryover-affected periods or includes a continuous drug-exposure regressor `Dbc` that absorbs exponential-decay carryover. This is a principled difference of analytical specification, not a disagreement on the underlying statistical behavior. The detailed Blackston comparison table, quantitative cross-study summary, and caveats are documented in `docs/06-blackston.md` §§1-7.

5.7.2 Chen, Chen, and Xia (2014): four analysis-method comparison

[87]

Chen, Chen, and Xia compared four analysis approaches across 3-cycle and 4-cycle aggregated N-of-1 trials, reporting Type I error control and power for compound-symmetry and AR(1) structures with and without carryover.

- Four analysis methods were compared: paired t-test, mixed differences model, full mixed model with carryover regressor, and DerSimonian-Laird meta-analysis.
- Three compound-symmetry covariance structures and two AR(1) structures were evaluated at $n \in \{1, 3, 5, 10, 20, 30\}$.

- Simulations used 3,000 replicates per cell with 0% or 20% carryover.

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Chen, Chen, and Xia[10] compared four analysis approaches: a paired t-test (M1), a mixed-effects model of differences (M2), a full mixed-effects model with carryover regressor (M3), and DerSimonian-Laird meta-analysis (M4), on simulated 3-cycle and 4-cycle aggregated N-of-1 trials at $n \in \{1, 3, 5, 10, 20, 30\}$ across three compound-symmetry covariance structures (covariances 0, 0.5, 0.8) and two AR(1) structures (coefficients 0.5, 0.8), with 0% or 20% carryover and 3,000 replicates per cell.

[88]

Under compound-symmetry structures, DerSimonian-Laird meta-analysis showed inflated Type I while other methods were near nominal; under carryover, only the mixed-effects model with explicit carryover term was unbiased.

- Under AR(1) structures, all models were slightly conservative (0.020 to 0.040).
- With 20% carryover, the M3 model with explicit carryover regressor was unbiased while M1, M2, and M4 power declined 1–10 percentage points.
- Chen and colleagues recommend the paired t-test as default and the mixed model with explicit carryover when carryover is present.

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Their Type I error findings under compound-symmetry structures showed M1 and M3 near nominal (0.031 to 0.061 across n at CS1); M2 conservative (0.013 to 0.044); and M4 inflated (0.036 to 0.081 across n at CS1). Under AR(1) structures, all models were slightly conservative (0.020 to 0.040). Power at effect 0.4, $n = 10$: M1 = 0.323 (CS1), 0.461 (CS2), 0.913 (CS3), 0.556 (AR1), 0.918 (AR2); M3 ran 5 to 30 percentage points below M1 in most cells. With 20% carryover, M3 power was essentially unchanged (because M3 includes the carryover regressor) while M1, M2, and M4 power declined 1 to 10 percentage points. Chen and colleagues recommend the paired t-test as the default and the mixed-effects model with explicit carryover term when carryover is present.

[89]

The present study extends Chen et al. by adding `corCAR1`

residual serial-correlation structure, and the findings are broadly concordant on Type I control and carryover handling.

- Chen et al.'s finding that Type I is controlled under 20% carryover by the M3 family is concordant with the S12 null calibration result.
- Chen et al.'s finding that DerSimonian-Laird meta-analysis performs poorly at small n motivated replacement of M4 with the M3-family `lme/corCAR1` model.
- Period exclusion and the continuous `Dbc` regressor represent complementary rather than competing analytical strategies.

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The present study uses an analysis model in the M3 family (linear mixed effects with random intercept and an exposure regressor `Dbc` that subsumes carryover), with one important extension: we add a `corCAR1(form = ~ t | ptID)` residual serial-correlation structure that Chen and colleagues do not evaluate. Their finding that Type I is controlled by M3 under 20% carryover is concordant with our S12 null calibration, where the hybrid maintains Type I in [0.056, 0.062] across carryover half-lives. Their finding that the paired t-test matches or exceeds mixed-effects power in most cells is not directly comparable because the paired t-test implicitly ignores carryover-affected periods (consistent with our period-exclusion analysis) while our hybrid uses the continuous `Dbc` regressor; both strategies exist in the design space for responding to carryover and are not in binary opposition. Chen and colleagues' finding that meta-analysis of summary measures (M4, DerSimonian-Laird on patient-level summaries) performs poorly at small n is worth noting as historical context: an earlier draft of this paper used M4 as the primary analysis model. The production simulation (`production-run.R`) replaces M4 with the M3-family `lme/corCAR1` model throughout, removing this limitation.

5.7.3 Tang and Landes (2020): serial t-tests for single-individual N-of-1

[90]

Tang and Landes developed serial t-tests correcting for AR(1) correlation at the single-individual level, demonstrating that uncorrected analyses inflate Type I and that power collapses under high positive serial correlation with short series.

- At $m = 12$, $\rho = 0.67$, the usual paired t-test achieves Type I of 0.22 while the serial t-test achieves 0.06.

- Effect sizes detectable with 80% power at $m = 4$ inflate from 1.65σ at $\rho = 0$ to 13.73σ at $\rho = 0.67$.
- These results demonstrate the critical importance of accounting for serial correlation in N-of-1 analyses.

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Tang and Landes[13] develop four analytical t-tests (paired-level, 2-sample-level, paired-rate, 2-sample-rate) that correct the usual Student's t-test for AR(1) serial correlation at the single-individual level. Their Monte Carlo simulation with 10,000 replicates per configuration reports Type I and power for m observations in $\{4, 5, \ldots, 12, 30, 50, 100\}$ at $\rho \in \{-0.33, 0, 0.33, 0.67\}$.

[91]

At $m = 12$, $\rho = 0.67$, the serial t-test restores near-nominal Type I error, and Table 2 of Tang and Landes quantifies the dramatic inflation of detectable effect sizes under positive serial correlation.

- The usual paired t-test achieves Type I of roughly 0.22 while the serial t-test achieves roughly 0.06.
- The detectable μ_{A-B} at $m = 4$, $\rho = 0.67$ is 13.73σ, a 48-fold inflation relative to the uncorrelated case.
- At $m = 12$, the same three autocorrelation levels yield detectable effects of 0.52σ, 1.25σ, and 5.43σ.

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At $m = 12$, $\rho = 0.67$, the usual paired t-test achieves a Type I rate of roughly 0.22, while the serial paired t-test achieves roughly 0.06, providing the central demonstration that uncorrected analyses inflate Type I under positive serial correlation. Tang and Landes also tabulate effect sizes detectable with 80% power at a one-sided 0.10 level (their Table 2): at $m = 4$, $\rho = 0$ the detectable μ_{A-B} is 1.65σ; at $\rho = 0.33$ it is 2.81σ; at $\rho = 0.67$ it is 13.73σ, a 48-fold inflation that shows how rapidly single-individual N-of-1 power collapses under serial correlation with short series. At $m = 12$, the same points are 0.52σ, 1.25σ, and 5.43σ.

[92]

The Tang and Landes results speak to single-subject analysis targets outside the aggregated N-of-1 scope; the aggregated hybrid absorbs within-patient correlation via `corCAR1`, producing concordant Type I behavior.

- The aggregated-level analogue finds hybrid power saturated across ρ because `corCAR1` absorbs within-patient correlation across 40 participants.

- Both papers predict that analyses ignoring positive residual correlation will inflate Type I.
- The extreme detectable-effect-size inflation at small m cautions against single-subject N-of-1 analysis with fewer than approximately ten observations.

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The Tang and Landes results speak to the single-subject analysis target that lies outside the aggregated N-of-1 scope of the present simulation. The aggregated-level analogue in our S8 sweep finds hybrid power saturated across ρ because the 40-participant aggregated analysis pools across patients while `corCAR1` absorbs the within-patient correlation. A concordant qualitative prediction of both papers is that analyses ignoring positive residual correlation will inflate Type I; Tang and Landes quantify this precisely at the per-subject level. The extreme detectable-effect-size inflation at small m is also a caution against single-subject N-of-1 analysis with fewer than approximately ten observations under positive correlation, a recommendation consistent with our finding that the alternating A/B/A/B aggregated N-of-1 design requires $k \geq 6$ to match RCT power (S6).

5.7.4 Concordance synthesis and remaining gaps

[93]

Across the five completed head-to-head comparisons, design ranking and direction of effects are concordant with the present results on all key dimensions.

- Aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson).
- Period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and Landes).
- Analyses ignoring carryover or serial correlation inflate Type I at the design level where those nuisance components operate (Blackston, Chen, Tang and Landes).

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Across the five completed head-to-head comparisons the design ranking and direction of effects are concordant with the present results: aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson); period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and

Landes); and analyses that ignore carryover or serial correlation inflate Type I at the design level at which those nuisance variance components operate (Blackston, Chen and colleagues, Tang and Landes). The quantitative specifics differ because the studies differ in target (main effect versus biomarker interaction), aggregation level (aggregated versus single-subject), parameterization (standardized τ versus nightmares-per-week effect), and analysis model (paired t-test versus mixed effects versus serial t-test versus `nlme::lme` with `corCAR1`).

[94]

Three prior-literature comparisons remain incomplete due to full-text unavailability, and are queued in the companion analysis document for future revision.

- Shi and Ye on behavioral carryover in 2-period crossovers is pending.
- Pequignat et al. on idiographic power for rare-condition precision medicine is pending.
- Chen, Li, and Yang on aggregated N-of-1 versus parallel and crossover RCTs is pending.

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Unfortunately, three comparisons remain incomplete: Shi and Ye[14] on behavioral carryover in 2-period crossovers, Pequignat and colleagues[48] on idiographic power for rare-condition precision medicine, and Chen, Li, and Yang[9] on aggregated N-of-1 versus parallel and crossover RCTs (a near-title-match with Blackston 2019). These are queued in the companion analysis (`docs/06-blackston.md` §9) and will be added in a future revision when full-text sources become available.

5.8 Future Research Directions

[95]

Empirical validation through head-to-head comparisons of N-of-1 and RCT designs in actual clinical trials would strengthen the evidence base for design selection.

- Real-trial comparisons would expose compliance and feasibility challenges absent from simulation.
- The simulation findings provide testable predictions that could be evaluated against actual trial data.
- Such validation is a natural next step for establishing the external validity of the simulation results.

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Our findings suggest several directions that warrant further investigation. First, empirical validation of these simulation results through head-to-head comparisons of N-of-1 and RCT designs in actual clinical trials would strengthen the evidence base for design selection[4].

[96]

The development of adaptive N-of-1 trial designs that optimize crossover periods based on accumulating data, particularly Bayesian adaptive designs, could further improve efficiency.

- Bayesian approaches are well-suited for adaptive N-of-1 designs.
- The simulation framework developed here could readily be extended to evaluate Bayesian adaptive variants.
- Such extensions would further address the joint n-versus-k optimization problem.

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Second, the development of adaptive N-of-1 trial designs that optimize the number of crossover periods based on accumulating data could further improve efficiency[8]. Bayesian approaches appear particularly well-suited for such adaptive designs, and the simulation framework developed here could readily be extended to evaluate them.

[97]

Investigation of hybrid designs combining N-of-1 and traditional RCT elements could capture advantages of both approaches while mitigating respective limitations.

- Hybrid designs may be particularly appropriate for settings where both individual and population-level inference are required.
- Quantitative evaluation of specific hybrid configurations can draw on the simulation framework developed here.
- Appropriate design selection depends on the relative weight given to individual versus population-level inferential goals.

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Third, investigation of hybrid designs that combine elements of N-of-1 and traditional RCT methodology could potentially capture the advantages of both approaches while mitigating their respective limitations[27].

[98]

Patient and clinician acceptance of N-of-1 methodology in real-world settings will be important for implementation, as treatment burden, complexity, and interpretability may influence practical utility.

- Factors such as treatment burden and complexity cannot be captured by simulation studies.
- Interpretability of individual versus population-level results may differ between clinical stakeholders.
- Implementation research complementing the present simulation evidence is warranted.

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Finally, examination of patient and clinician acceptance of N-of-1 trial methodology in real-world settings will be important for successful implementation[26]. Factors such as treatment burden, complexity of implementation, and interpretability of results may influence the practical utility of N-of-1 approaches in ways that simulation studies cannot capture.

5.9 Preliminary validation run

[99]

Before the production simulation was finalized, a preliminary validation run on the same 8-cell design slice verified computational feasibility and confirmed the direction of the main findings, while identifying two implementation issues.

- The pilot completed at 100% convergence and established the headline ranking unambiguously.
- Two implementation issues were identified: undercalibrated within-patient residual SD and failure to exclude carryover-contaminated periods.
- Both issues were corrected in the production run.

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Before the production simulation was finalized, we ran a preliminary validation on the same 8-cell design slice (design $\times$ effect-size grid) to verify computational feasibility and confirm the direction of the main findings. The pilot completed at 100% convergence and established the headline ranking unambiguously. It also identified two implementation issues.

[100]

The within-patient residual SD in the initial run was too small relative to the target the mean-moderation architecture Gompertz DGP, yielding empirical SE approximately 0.107 versus the target 0.28.

- The production run addresses this by calibrating `SIGMA_WITHIN` to 2.3.
- This calibration yields empirical SE approximately 0.28 at the null cell.
- The undercalibrated SD produced artificially narrow effect-estimate distributions, inflating apparent power in the pilot.

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First, the within-patient residual standard deviation in the initial run was set too small relative to the target the mean-moderation architecture Gompertz DGP (empirical SE ≈ 0.107 versus target ≈ 0.28). The production run addresses this by calibrating `SIGMA_WITHIN` to 2.3, yielding empirical SE ≈ 0.28 at the null cell.

[101]

The initial run did not exclude carryover-contaminated placebo periods, causing bias of $\hat{\beta}_D$ to grow with $|\beta_D|$ and coverage to collapse from 94.9% at the null to 37.5% at $\beta_D = -2$.

- The production run excludes contaminated periods using the `is_carryover` flag set by the data-generating process.
- Exclusion removes the bias and restores coverage to its nominal level.
- This finding confirms that period exclusion is necessary for unbiased estimation under the present DGP.

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Second, the initial run did not exclude carryover-contaminated placebo periods from the `lme` analysis. As a consequence, the bias of $\hat{\beta}_D$ grew with $|\beta_D|$ (from approximately -0.20 at the null to -2.12 at $\beta_D = -2$), and coverage collapsed from 94.9% at the null to 37.5% at $\beta_D = -2$. The production run excludes contaminated periods before fitting (using the `is_carryover` flag set by the data-generating process), which removes the bias and restores coverage to its nominal level.

6 Conclusions

[102]

We have reported a simulation study comparing aggregated N-of-1 hybrid trials and parallel-group RCTs at matched participant count across a clinically calibrated range of effect sizes and carryover half-lives.

- Three principal conclusions are emphasized.
- The comparison is conducted under the ADEMP framework with Monte Carlo standard errors for all performance measures.
- Parameters are calibrated to the prazosin/PTSD reference application.

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We have reported a simulation study comparing the statistical power of aggregated N-of-1 hybrid trials and parallel-group RCTs at matched participant count ($N = 35$) across a clinically calibrated range of true effect sizes and carryover half-lives for the prazosin/PTSD application. In conclusion, three points are to be emphasized.

[103]

The N-of-1 hybrid achieved substantially higher power than the parallel-group RCT at every non-null effect size, with the largest advantage in the small-to-moderate effect regime where the parallel RCT reaches only 10–23% power.

- The N-of-1 hybrid achieved 31–83% power at $\beta_D \in [-1.0, -0.5]$ nightmares per week versus 10–23% for the RCT.
- The absolute power gap exceeded five combined MCSEs at every non-null cell.
- The ranking is not attributable to simulation noise.

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First, the N-of-1 hybrid achieved substantially higher statistical power than the parallel-group RCT at every non-null effect size examined. The power advantage was largest in the small-to-moderate effect regime ($\beta_D \in [-1.0, -0.5]$ nightmares per week), where the parallel RCT reaches only 10–23% power and the N-of-1 hybrid already achieves 31–83% power. The absolute gap exceeded five combined MCSEs at every non-null cell, confirming that the ranking is not attributable to simulation noise.

[104]

Period exclusion for carryover management proved robust: power, bias, and empirical SE were essentially constant across all five carryover half-life cells, and bias was negligible with coverage tracking the nominal 95% level throughout.

- The fitted model operates only on non-contaminated periods after exclusion, making the true half-life irrelevant to the estimating equations.
- Robustness held across $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days in the S1 sweep.
- Coverage was maintained at the nominal level for both designs at all effect-size cells.

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Second, the period-exclusion strategy for carryover management proved robust: power, bias, and empirical SE were essentially constant across all five carryover half-life cells in the S1 sweep ($t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days), because the fitted model operates only on the non-contaminated periods. Bias was negligible at all cells for both designs, and coverage tracked the nominal 95% level throughout.

[105]

The `lme`/`corCAR1` model with Satterthwaite denominator degrees of freedom provided adequate Type I error control at $N = 35$; the parallel-group RCT showed a marginally elevated rate warranting further investigation.

- N-of-1 Type I error was 0.051, within one MCSE of the nominal 0.05.
- RCT Type I error was 0.061, consistent with known small-sample effects in `lm` at $N = 35$.
- The marginally elevated RCT rate warrants further investigation before the final manuscript is submitted.

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Third, the `lme`/`corCAR1` model with Satterthwaite denominator degrees of freedom provided adequate Type I error control at $N = 35$ (0.051 for N-of-1, within one MCSE of the nominal 0.05). The parallel-group RCT showed a marginally elevated Type I rate of 0.061, consistent with known small-sample effects in lm at $N = 35$, which warrants further investigation before the final manuscript is submitted.

7 Acknowledgments

[106]

We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for

simulation parameters, and acknowledge the methodological contributions of the N-of-1 trials research community.

- Empirical parameter calibration relied on published prazosin/PTSD clinical trials.
- Analytical frameworks from the N-of-1 methodology literature provided the foundation for the simulation design.
- The N-of-1 research community's ongoing methodological contributions are acknowledged.

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We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for our simulation parameters, and acknowledge the methodological contributions of the N-of-1 trials research community in developing the analytical frameworks on which this work draws.

8 Author Contributions

[To be filled in based on actual authorship]

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

[107]

Simulation data and code supporting the findings are openly available in the pmsimstats-ng project repository, with per-replicate Monte Carlo outputs and ADEMP-compliant summaries versioned under `analysis/data/`.

- Per-replicate Monte Carlo outputs are versioned under `analysis/data/`.
- Simulation drivers are located at `analysis/scripts/`.
- Exact paths for this manuscript are listed in the Reproducibility section.

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

[108]

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container, with package dependencies captured in `renv.lock` and the principal simulation packages listed.

- The zzcollab Docker container image hash is recorded in the `Dockerfile`.
- Principal packages include `nlme`, `parallel`, `data.table`, `furrr`, `purrr`, `rmarkdown`, and `bookdown`.
- Full reproducibility requires both the container image and the `renv.lock` package manifest.

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[109]

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`, with per-replicate seeds derived from the master seed.

- The master seed is applied once at program entry.
- Per-replicate seeds derive from the master seed by `+ rep_idx`.
- This scheme permits exact reproduction of any replicate while maintaining independence across replicates.

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Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[110]

The canonical production simulation output is at `analysis/data/derived_data/04-mainfx-prod` and the driver scripts, ADEMP plan, and manuscript source are fully documented.

- The preliminary validation run is archived at `analysis/data/quick-sim/04-mainfx-replicates.r`
- Driver scripts are under `analysis/scripts/treatment-main-effect/`; the production driver is `production-run.R`.
- The pre-registered ADEMP plan is at `analysis/scripts/treatment-main-effect/00-ademp-pre-`

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Data and code availability. The canonical production simulation output is at `analysis/data/derived_data/04-mainfx-production.rds`. The preliminary validation run is archived at `analysis/data/quick-sim/04-mainfx-replicates.rds`. Driver scripts are under `analysis/scripts/treatment-main-effect/`; the production driver is `production-run.R`. The pre-registered ADEMP plan is at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/04-treatment-main-effect/`.

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14 Supporting Information

S1 File. Simulation code and reproducibility materials. Complete R code for all simulations, including parameter specifications, statistical analysis functions, and visualization code. The simulation framework includes Morris et al. (2019) compliant functions for calculating Monte Carlo standard errors, Type I error evaluation, and comprehensive performance summaries.

S2 File. Extended sensitivity analyses. Additional simulation results examining robustness to alternative parameter specifications and design variations. Includes comparison of carryover handling strategies (period exclusion versus explicit carryover modeling) and carryover decay model sensitivity analysis (exponential, linear, and Weibull decay functions).

S3 File. Individual N-of-1 trial examples. Detailed illustrations of individual crossover patterns and analytical approaches for representative patients.

S4 File. Monte Carlo standard error calculations. Technical documentation of MCSE formulas following Morris et al. (2019) Table 6, including formulas for power, bias, empirical SE, MSE, and coverage.

Competing Interests: The authors have declared that no competing interests exist.

Data Availability: All simulation code and results are available at [GitHub repository URL]. No human subjects data were collected for this study.

Funding: [To be filled in based on actual funding sources]

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